Efficient Estimation of Continuous Treatment Effects

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February, 2026

Abstract

Estimating causal effects of continuous treatments is challenging because traditional methods rely on unstable estimation of the score of the conditional treatment density. We address this instability by extending the balancing weights framework to continuous treatments. By shifting the analytical primitive to the outcome weight, we guarantee a bounded statistical functional by design and bypass density derivative estimation. Within this framework, we characterize the nonparametric efficiency bound and derive the optimal continuous balancing weights under homoskedasticity and heteroskedasticity. Under homoskedasticity, the optimally efficient estimand coincides with the population projection coefficient in a Partially Linear Regression. Under heteroskedasticity, the optimal estimands center the treatment around a precision-weighted propensity score, generalizing the overlap-weighted effect of categorical treatments. Finally, we derive the efficient influence functions and develop $\sqrt{n}$-consistent estimators based on Double/Debiased Machine Learning, providing a semiparametric toolkit for empirical practice. We illustrate the proposed estimators in an application regarding the effect of winning a lottery prize on labor supply.

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1 Introduction

Estimating the causal effects of continuous treatments or exposures, such as drug dosage, the intensity of a policy intervention, or exposure to economic shocks, is a fundamental challenge in empirical economics. While the dose-response curve, $d \mapsto \mathbb{E}[Y_i(d)]$, provides a complete characterization of the treatment effect of dose d, flexibly estimating this entire function is a statistically demanding task typically characterized by slow, nonparametric convergence rates. In many empirical settings, researchers require a robust scalar summary of the marginal effect that can be estimated at the parametric root-n rate.

For continuous treatments, a natural scalar summary is a weighted Average Derivative Effect (ADE), taking the form $\tau_w = \mathbb{E}[w(D_i, X_i)\mu'(D_i, X_i)]$, where D_i is the treatment and X_i is the set of covariates, $\mu(d, x) = \mathbb{E}[Y_i \mid D_i = d, X_i = x]$ is the conditional expectation function, μ' is its partial derivative with respect to d, and $w(d, x)$ is a user-specified weighting function defining the target population. However, the efficient estimation of the ADE also presents statistical challenges. From a functional analysis perspective, the target parameter τ_w evaluates a differential operator applied to the unknown regression function μ. Consequently, the mapping $\mu \mapsto \mathbb{E}[w\mu']$ is generally an unbounded linear functional, and thus, one cannot directly invoke the Riesz representation theorem in $L_2(P_{D,X})$ to guarantee the existence of a stable, square-integrable representer.

Instead, traditional semiparametric estimation must construct a representation explicitly. This is typically achieved via integration by parts, which shifts the unbounded derivative operator from the unknown conditional mean μ to the conditional density of the treatment $f(d|x)$ (Powell et al., 1989; Newey and Stoker, 1993). This operation yields a weighting function on the observed outcome, $\alpha_w(d, x) = -\partial_d w(d, x) - w(d, x)\partial_d \log f(d|x)$, such that $\tau_w = \mathbb{E}[\alpha_w(D_i, X_i)Y_i]$. This method, while specifying the target population w and subsequently deriving its representer α_w, exposes a theoretical vulnerability. The explicitly constructed α_w inherently relies on the score of the conditional density, $\partial_d \log f(d|x)$. Nonparametrically estimating the derivative of a conditional density is an ill-posed inverse problem. It amplifies finite-sample noise, requires strong smoothness assumptions, and suffers from severe instability when the density is near zero (e.g., Newey and McFadden, 1994; Cattaneo et al., 2010).

This instability mirrors a well-known vulnerability in the categorical treatment literature, where standard Average Treatment Effect (ATE) estimators rely on inverse probability weights that blow up when propensity scores approach the boundaries. To resolve the limited overlap problem, researchers developed the "balancing weights" framework (Crump et al., 2006; Li et al., 2018; Li and Li, 2019). By shifting the estimand to prioritize subpopulations with substantial covariate overlap, this framework directly parameterizes weights that minimize the asymptotic variance of the estimator, thereby identifying inherently stable estimands and bypassing extreme propensity weights.

Recent work have extended covariate balancing principles to continuous treatments. For example, Fong et al. (2018) and Ai et al. (2021) propose unified frameworks that directly estimate continuous stabilized weights, $f(d)/f(d|x)$, by forcing covariates and treatments to be uncorrelated. While these methods bypass maximum likelihood density estimation, they are designed to recover causal dose-response levels $\mathbb{E}[Y(d)]$. When researchers require a summary of the marginal effect, these level-based IPW frameworks generally restrict analysis to parameterized structural models (e.g., linear regressions) or discrete contrasts (e.g., $\mathbb{E}[Y(d') - Y(d)]$), instead of a fully nonparametric summary of the derivative effects.

In this paper, we overcome the instability of continuous treatment effects by extending the balancing weights framework to the continuous setting. Our key innovation lies in reversing the traditional analytical paradigm. Rather than specifying a derivative weight w and employing an unbounded differential operator to derive an unstable representer α_w, we propose defining the target estimand by directly specifying a well-behaved, square-integrable outcome weight $\alpha \in L_2(P_{D,X})$. By restricting the definition to a bounded α from the beginning, we guarantee that the mapping $\mu \mapsto \mathbb{E}[\alpha\mu]$ is a bounded, continuous linear functional by design.

In independent and concurrent work in the statistics literature, Hines et al. (2023) explore a similar class of continuous estimands. However, they implicitly assume that the derivative mapping $\mu \mapsto \mathbb{E}[w\mu']$ is inherently a bounded linear functional. We highlight that this differential operator is inherently unbounded, and by diagnosing this unboundedness, we prove that defining the outcome weight α as the analytical primitive is not merely a statistical convenience, but a necessity to guarantee a well-posed functional.

Furthermore, our analysis is embedded within the causal inference framework. For the statistical functional $\mathbb{E}[\alpha\mu]$ to identify a causally interpretable average derivative free from confounding bias, α must be structurally orthogonal to the pre-treatment covariates, i.e.,

$\mathbb{E}[\alpha(D_i, X_i) \mid X_i] = 0$. This orthogonality condition serves as the continuous treatment generalization of the covariate balance required in the categorical treatment literature (Li et al., 2018). Crucially, by treating α as the primitive, the implied derivative weight, $w_\alpha(d, x)$, describes the aggregation scheme of the conditional causal derivatives. This contributes directly to an active and growing econometrics literature evaluating implicit treatment effect weights in regressions (e.g., Blandhol et al., 2022; Goldsmith-Pinkham et al., 2022; Borusyak and Hull, 2024).

Finally, we seek the optimally efficient way to summarize the continuous treatment effect and derive the optimal balancing weight α^* that minimizes the nonparametric efficiency bound among the class of all valid average derivative estimands, accommodating homoskedastic and heteroskedastic cases. We discover that, under homoskedasticity, the optimally efficient continuous estimand τ^* corresponds exactly to the population projection coefficient in a Partially Linear Regression (PLR) (Robinson, 1988). This equivalence result formally provides a causal interpretation of the PLR coefficient and a efficiency-based justification for its use in empirical research. Under heteroskedasticity, the optimal estimands are novel contributions. We then derive the Efficient Influence Functions of the optimal estimands, and develop semiparametric estimators which are consistent and asymptotically normal based on the Double Machine Learning (DML) framework. While Hines et al. (2023) also obtained similar optimality results, they focus on the estimation of the projection parameters in partially linear regressions.

The remainder of the paper is organized as follows. Section 2 introduces the functional analysis framework, formulates generalized balancing weights for continuous treatments. Section 3 analyzes the nonparametric efficiency bounds, derives the optimal balancing weights and the corresponding estimands. Section 4 discusses the estimation of the optimal estimands and establishes their asymptotic properties. Section 6 illustrates the optimal estimand through an empirical application examining the effect of lottery winnings on labor supply. Section 7 concludes. All proofs are collected in the Appendix.

2 Balancing Weights for Continuous Treatments

Suppose we observe an independent and identically distributed sample of n observations $O_i = (Y_i, D_i, X_i)$ drawn from an unknown joint distribution P. Here, $Y_i \in \mathbb{R}$ is the observed outcome, $D_i \in \mathcal{D} \subseteq \mathbb{R}$ is a continuous treatment or exposure, and $X_i \in \mathbb{R}^p$ is a vector of pre-treatment covariates. We adopt the potential outcomes framework, letting $Y_i(d)$ denote the potential outcome for unit i had they been assigned to treatment level $d \in \mathcal{D}$.

While the dose-response curve $d \mapsto \mathbb{E}[Y_i(d)]$ provides a full characterization of the treatment effect, estimating this entire function nonparametrically is a statistically demanding task that suffers from the curse of dimensionality. In many empirical settings, researchers require a robust scalar summary of the marginal effect that can be estimated at the parametric $n^{-1/2}$ rate.

To formalize a causal target parameter, we define the weighted average causal derivative as the limit of an expected incremental policy shift

$$\dot{\tau}_w = \lim_{\nu \to 0} \mathbb{E}\left[w(D_i, X_i) \frac{Y_i(D_i + \nu) - Y_i(D_i)}{\nu} \right],$$

where $w(d, x)$ is a weighting function. The causal parameter $\dot{\tau}_w$ characterizes the effect of increasing the treatment level by ν on the expected potential outcome (e.g., Kreif et al., 2015)

2.1 Identification

To connect the causal parameter $\dot{\tau}_w$ to the observable data distribution P, let $\mu(d, x) = \mathbb{E}[Y_i \mid D_i = d, X_i = x]$ denote the observed conditional expectation function, with its partial derivative with respect to d denoted as $\mu'(d, x) = \partial_d \mu(d, x)$. Let $f(d|x)$ denote the conditional density of D_i given $X_i = x$. We define the conditional expected potential outcome as $m(d, x) = \mathbb{E}[Y_i(d) \mid X_i = x]$. We formalize causal identification under the following assumptions.

Assumption 1. *(a) $D_i = d$ implies $Y_i = Y_i(d)$.*

(b) $\mathbb{E}[Y_i(d) \mid D_i, X_i] = \mathbb{E}[Y_i(d) \mid X_i]$ for all $d \in \mathcal{D}$.

(c) For any point (d_0, x_0) in the support of P, $f(d|x_0)$ is continuous in d and strictly positive in an open neighborhood of d_0.

(d) The conditional expected potential outcome $m(d, x)$ is continuously differentiable with respect to d, and its derivative $\partial_d m(d, x)$ is uniformly bounded.

(e) $\mathbb{E}[|w(D_i, X_i)|] < \infty$.

Assumption 1 adapts traditional causal requirements to the continuous setting with maximal flexibility. Assumption 1(a) is the SUTVA assumption. Assumption 1(b) utilizes a weak version of unconfoundedness, requiring only conditional mean independence rather than full distributional independence. Assumption 1(c) is a localized version of the overlap assumption, which ensures that the conditional mean $\mu(d, x)$ is well-defined in an open neighborhood around any observed treatment point, avoiding extreme extrapolation. Assumption 1(d) specifies differentiability requirement on the conditional average expectation $m(\cdot, x)$. Assumption 1(e) bounds the weight function $w(d, x)$. Under these conditions, the average causal derivative is nonparametrically identified by the average derivative of the observed conditional mean.

Proposition 1. *Under Assumption 1, the weighted average causal derivative is identified by the observable statistical functional*

$$\dot{\tau}_w = \mathbb{E}[w(D_i, X_i)\mu'(D_i, X_i)].$$

In the proof, we show that under Assumption d, the derivative $Y_i'(d)$ represents the limiting incremental effect as the treatment increment approaches zero, formally expressed as $\mathbb{E}[Y_i'(d)] = \lim_{\nu \to 0} \frac{\mathbb{E}[Y_i(d+\nu)] - \mathbb{E}[Y_i(d)]}{\nu}$. This relationship allows us to interpret the weighted average derivative as a causal parameter based on incremental effects. Furthermore, it enables the aggregation of derivative effects across treatment levels through integration of $\mathbb{E}[Y_i(d)|X_i = x]$ over $f(d|x)$. When $D_i \in \{0, 1\}$, the analogous identification result for binary treatment is $\mathbb{E}[w(X_i)(Y_i(1) - Y_i(0))] = \mathbb{E}[w(X_i)(\mu(1, X_i) - \mu(0, X_i))]$, which is developed in the balancing weights literature (Crump et al., 2006; Li et al., 2018).

In light of Proposition 1, we drop the dot notation for the remainder of the paper and focus on the statistical estimand $\tau_w \equiv \mathbb{E}[w(D_i, X_i)\mu'(D_i, X_i)]$, keeping in mind its structural justification as the continuous limit of a causal shift intervention.

2.2 The Instability of Average Derivatives

Let $\mathcal{H} = L_2(P_{D,X})$ denote the Hilbert space of all measurable functions $g : \mathcal{D} \times \mathbb{R}^p \to \mathbb{R}$ that are square-integrable with respect to the joint distribution of (D_i, X_i). The parameter τ_w

evaluates a linear functional of the conditional response surface

$$T_w(\mu) = \mathbb{E}[w(D_i, X_i)\mu'(D_i, X_i)].$$

In modern semiparametric theory, achieving root-n consistent estimation of a linear functional fundamentally relies on finding its Riesz representer, a function $\alpha_w \in \mathcal{H}$ such that $T_w(\mu) = \langle \alpha_w, \mu \rangle = \mathbb{E}[\alpha_w(D_i, X_i)\mu(D_i, X_i)]$. By the law of iterated expectations, the existence of such an α_w allows the estimand to be expressed directly as a weighted average of the observed outcome $\tau_w = \mathbb{E}[\alpha_w(D_i, X_i)Y_i]$.

However, a mathematical obstacle arises. The functional T_w applies a differential operator, $\mu \mapsto \mu'$. In the standard $L_2(P_{D,X})$ topology, differential operators are strictly unbounded linear functionals. While Hines et al. (2023) assumes this mapping is bounded to justify algorithmic convenience, standard functional analysis shows that one cannot invoke the Riesz Representation Theorem to guarantee the existence of a unique, square-integrable representer for an unbounded operator.

Because the Riesz Representation Theorem does not apply, traditional econometric approaches construct a representation explicitly. Under specific boundary and density smoothness assumptions (detailed as Assumption 3 in Appendix A), one can apply integration by parts to shift the derivative operator from the unknown outcome function μ to the conditional treatment density $f(d|x)$ (Powell et al., 1989),

$$\alpha_w(d, x) = -\frac{\partial w(d, x)}{\partial d} - w(d, x)\frac{\partial \log f(d|x)}{\partial d}. \tag{1}$$

If the weighting scheme w and the underlying data-generating process are sufficiently regular to ensure this explicitly constructed α_w has finite variance ($\alpha_w \in \mathcal{H}$), the functional is bounded ex post, and α_w algebraically serves as its representer.

Equation (1) illustrates the theoretical vulnerability of estimating continuous treatment effects. Because the original functional is unbounded, deriving its representer forces a structural reliance on the score of the conditional density, $\partial_d \log f(d|x)$. Consequently, the traditional analytical method, which fixes the derivative weight w as the primitive and subsequently maps $w \mapsto \alpha_w$, demands restrictive smoothness conditions on the unobservable data-generating process. Even when these conditions hold, nonparametrically estimating the derivative of a conditional density is an ill-posed inverse problem. It amplifies finite sample noise and causes estimators to become unstable when $f(d|x)$ is close to zero, acting as a continuous analogue to the limited overlap problem in binary treatments (e.g., Cattaneo et al., 2010).

2.3 Generalized Balancing Weights

The instability inherent in Equation (1) arises because the standard approach specifies the target population via the derivative weight w, and then employs an unbounded differential operator to derive the representer α_w. We propose to invert this analytical sequence. Rather than specifying w, we define the statistical estimand by directly parameterizing an outcome weight $\alpha \in \mathcal{H}$ as our analytical primitive.

By restricting our primitive α to bounded elements in $\mathcal{H}$ from the beginning, the resulting linear functional $T_\alpha(\mu) = \mathbb{E}[\alpha(D_i, X_i)\mu(D_i, X_i)]$ is guaranteed to be bounded and continuous by the Cauchy-Schwarz inequality ($|T_\alpha(\mu)| \leq \|\alpha\|_{\mathcal{H}}\|\mu\|_{\mathcal{H}}$). Under this inverted formulation, α serves as the Riesz representer by design.

However, for this bounded functional $\mathbb{E}[\alpha(D_i, X_i)Y_i]$ to identify a causally meaningful average derivative effect, α cannot be an arbitrary function in $L_2(P_{D,X})$. It must satisfy econometric constraints to purge confounding bias and correctly scale the treatment variation. We formalize these requirements in the following definition.

Definition 1 (Continuous Balancing Weights). *A function $\alpha \in L_2(P_{D,X})$ is defined as a continuous balancing weight if it satisfies two structural moment conditions: (i) Covariate Balance: $\mathbb{E}[\alpha(D_i, X_i) \mid X_i] = 0$ almost surely; (ii) Normalization: $\mathbb{E}[\alpha(D_i, X_i)D_i] = 1$.*

Condition (i) serves a dual purpose. Statistically, it ensures that the boundary terms vanish during integration by parts, isolating the pure derivative effect. Causally, it generalizes the concept of covariate balance established in the categorical treatment literature. For categorical treatments, balancing weights are defined by their ability to equalize the weighted covariate distributions across treatment regimes (Li et al., 2018). Condition (i) ensures this distributional covariate balance across a continuum of treatment levels. Condition (ii) ensures the estimand correctly captures the magnitude of the marginal effect, which implies $\mathbb{E}[w(D_i, X_i)] = 1$. We formally show these arguments in the proof of the following proposition.

Let $F(d|x)$ denote the conditional cumulative distribution function (CDF) of D_i given $X_i = x$.

Proposition 2. *For any continuous balancing weight in Definition 1, the statistical estimand* $\tau_\alpha = \mathbb{E}[\alpha(D_i, X_i)Y_i]$ *identically recovers a normalized weighted average causal derivative, i.e.,* $\mathbb{E}[w_\alpha(D_i, X_i)] = 1$, *with the implied derivative weight uniquely defined as*

$$w_\alpha(d, x) = -\frac{F(d|x)}{f(d|x)} \cdot \mathbb{E}\big[\alpha(D_i, X_i) \mid D_i \leq d, X_i = x\big]. \tag{2}$$

Finally, a standard requirement for causal interpretation is that the implied derivative weights are non-negative. Non-negative weighting prevents the artificial "sign reversal" of local treatment effects, related to recent econometric discussion on design-based regressions (Blandhol et al., 2022; Goldsmith-Pinkham et al., 2022; Borusyak and Hull, 2024). Proposition 2 allows us to characterize the necessary and sufficient condition on the balancing weight α to guarantee non-negative $w_\alpha(d, x)$.

Lemma 1. *For a continuous balancing weight* α *and its implied weight* w_α *in Equation (2),* $w_\alpha(d, x) \geq 0$ *if and only if* $\mathbb{E}[\alpha(D_i, X_i) \mid D_i \leq d, X_i = x] \leq 0$ *for all* d *and* x.

Intuitively, for the truncated expectation to be non-positive at the lower boundary of the treatment support, $\alpha(d, x)$ must generally take negative values for small d. Conversely, for the overall expectation $\mathbb{E}[\alpha(D_i, X_i) \mid X_i]$ to return to exactly zero at the upper boundary, $\alpha(d, x)$ must transition to positive values for large d. Thus, Lemma 1 enforces a structural requirement that a valid causal balancing weight $\alpha(d, x)$ must transition from negative to positive as treatment intensity increases. While this condition provides an explicit diagnostic check for weight convexity, we treat the conditions in Definition 1 as the sole restriction in Section 3, as the variance minimizing solution we develop inherently adheres to Lemma 1.

3 Efficiency Bounds and Optimal Balancing Weights

3.1 Optimization Criterion and Regularity Conditions

The class of balancing weights is large, and Proposition 2 confirms that any continuous balancing weight identifies a normalized effect τ_w. This raises a crucial question: which balancing weight should be selected? Following the established approach in the balancing weights literature (Crump et al., 2006; Li and Li, 2019), we seek the balancing weight that is optimal in the sense of minimizing the nonparametric efficiency bound.

The efficiency bound characterizes the lowest possible asymptotic variance achievable by any regular asymptotically linear (RAL) estimator. When the derivative weight $w(d, x)$ is known, the influence function of T_w has been derived by Newey and Stoker (1993):

$$\psi(O_i) = \underbrace{\alpha_w(D_i, X_i)\big(Y_i - \mu(D_i, X_i)\big)}_{\equiv \psi_A(O_i)} + \underbrace{w(D_i, X_i)\mu'(D_i, X_i) - \tau_w}_{\equiv \psi_B(O_i)}. \tag{3}$$

Given the influence function and an efficient estimator $\hat{\tau}_w$, its asymptotic variance is given by

$$V_\psi = \mathbb{E}[\psi^2(O_i)] = \mathbb{E}[\psi_A^2(O_i)] + \mathbb{E}[\psi_B^2(O_i)], \tag{4}$$

as $\mathbb{E}[\psi_A(O_i)\psi_B(O_i)] = 0$, which is a consequence of the orthogonality between the residual term $Y_i - \mu(D_i, X_i)$ and the conditional derivative effects. Here, $\mathbb{E}[\psi_A^2(O_i)]$ represents the variance contribution from the noise in the outcome amplified by the magnitude of the balancing weights α_w. $\mathbb{E}[\psi_B^2(O_i)]$ is a measure of effect heterogeneity, representing the variance of the weighted individual derivative effects around the population average τ_w.

While minimizing the total variance V_ψ might seem intuitive, selecting w by minimizing the heterogeneity component $\mathbb{E}[\psi_B^2(O_i)]$ is conceptually and practically problematic. Because $\mathbb{E}[\psi_B^2(O_i)]$ is a function of the unknown population parameter τ_w, optimizing it leads to a circular definition. Furthermore, minimizing effect heterogeneity changes the objective from seeking statistical precision to contorting the targeted causal parameter based on the variation in treatment effects.

To optimize solely for statistical efficiency, we follow the strategy in Crump et al. (2006) and redefine the objective as minimizing the asymptotic variance of the estimator relative to the sample analogue of τ_w. The asymptotic variance of $\sqrt{n}(\hat{\tau}_w - \tau_{w,S})$ is the expectation of the square of the leading term $\psi_A(O_i)$, which gives the conditional efficiency bound:

$$V_S = \mathbb{E}[\psi_A^2(O_i)] = \mathbb{E}[\alpha_w^2(D_i, X_i)\big(Y_i - \mu(D_i, X_i)\big)^2]. \tag{5}$$

Let $\sigma^2(D_i, X_i) = Var(Y_i \mid D_i, X_i)$ denote the conditional variance of the outcome. By the law of iterated expectations, the objective function in Equation (5) simplifies cleanly to

$V_S = \mathbb{E}[\alpha_w^2(D_i, X_i)\sigma^2(D_i, X_i)]$.

The solution to this functional optimization problem depends on the form of heteroskedasticity. To formalize a unified framework across all heteroskedastic architectures, we define the generalized precision weight as

$$\omega(D_i, X_i) \equiv \sigma^{-2}(D_i, X_i). \tag{6}$$

To ensure the optimization is well-defined and that the resulting optimal estimands possess square-integrable influence functions, we assume the following regularity conditions.

Assumption 2. *(a) There exist strictly positive constants c_1, c_2 such that $c_1 \leq Var(Y_i \mid D_i, X_i) \leq c_2$ almost surely. Consequently, the precision weight is bounded by $1/c_2 \leq \omega(D_i, X_i) \leq 1/c_1$.*

(b) There exists $c_3 > 0$ such that $\mathbb{E}\big[Var(D_i \mid X_i)\big] \geq c_3$.

Remark 1. Assumption 2 imposes regularity conditions that are complementary to the causal identification conditions in Assumption 1. First, while Assumption 1(c) imposes a local positivity requirement on the conditional density $f(d|x)$ to ensure the causal derivative is nonparametrically identifiable point-by-point, Assumption 2(b) imposes a global requirement that the precision-weighted conditional variance of the treatment is strictly bounded away from zero. This guarantees that the denominator of the optimal estimand does not degenerate asymptotically. Second, Assumption 1(e) imposes $\mathbb{E}[|w(D_i, X_i)|] < \infty$ on derivative weights. For the optimal estimands derived in this section, Assumptions 2(a) and 2(c) provide the primitive conditions that guarantee the optimal balancing weights have finite variance. Finally, note that Assumption 2(c) is a condition for both deriving the optimal estimands and developing $\sqrt{n}$-consistent estimators, however, for the former, it can be relaxed to $q \geq 4$.

3.2 Optimal Continuous Balancing Weights

We select the optimal balancing weight α^* as the unique function that minimizes the conditional variance bound V_S subject to the balancing constraints in Definition 1.

By solving this constrained functional optimization problem in $L_2(P_{D,X})$, we obtain a unified theorem that cleanly characterizes the optimal estimands across different heteroskedasticity architectures. To express the general solution elegantly, we define the precision-weighted propensity as

$$e_\omega(x) = \frac{\mathbb{E}[\omega(D_i, X_i) D_i \mid X_i = x]}{\mathbb{E}[\omega(D_i, X_i) \mid X_i = x]}. \tag{7}$$

Note that when the outcome variance is conditionally independent of the treatment, $e_\omega(x)$ collapses to the standard propensity $e(x) = \mathbb{E}[D_i \mid X_i = x]$.

Theorem 1 (Optimal Continuous Balancing Weights)**.** *Suppose Assumption 2 holds. The optimal balancing weight α^* that minimizes the efficiency bound V_S, and its corresponding optimal estimand τ^*, are uniquely determined by the structure of the precision weight ω*

$$\alpha^*_\omega(d, x) = \frac{\omega(d, x)\big(d - e_\omega(x)\big)}{\mathbb{E}\big[\omega(D_i, X_i)\big(D_i - e_\omega(X_i)\big)^2\big]}, \tag{8}$$

$$\tau^*_\omega = \frac{\mathbb{E}\big[\omega(D_i, X_i)\big(D_i - e_\omega(X_i)\big)Y_i\big]}{\mathbb{E}\big[\omega(D_i, X_i)\big(D_i - e_\omega(X_i)\big)^2\big]}. \tag{9}$$

(a) General Heteroskedasticity: If $\sigma^2(D_i, X_i)$ varies with both D_i and X_i, the precision weight is $\omega(d, x) = \sigma^{-2}(d, x)$.

(b) Covariate-Dependent Heteroskedasticity: If the conditional variance depends only on pre-treatment covariates, $\omega(d, x) = \sigma^{-2}(x)$. The precision-weighted propensity simplifies to $e(x)$.

(c) Homoskedasticity: If the outcome is homoskedastic, $\omega(d, x) = 1$. The optimal balancing weight reduces to $\alpha^*(d, x) = (d - e(x))/\mathbb{E}[Var(D_i \mid X_i)]$, yielding the optimal estimand $\tau^* = \mathbb{E}[Cov(D_i, Y_i \mid X_i)]/\mathbb{E}[Var(D_i \mid X_i)]$.

Theorem 1 provides a characterization of efficiency in continuous treatment settings, yielding three key insights. First, under general heteroskedasticity, the structural noise level fluctuates along the continuous treatment path itself. In this scenario, orthogonalizing against the standard propensity $e(X_i)$ is no longer theoretically optimal. Instead, efficiency dictates that the treatment distribution must be centered around $e_\omega(X_i)$, which serves as the optimal predictor of D_i under a precision-weighted inner product. This localized centering guarantees that the estimand heavily weights the specific segments of the dose-response curve that exhibit the highest signal-to-noise ratio.

Crucially, this continuous optimization framework unifies established optimal weights from the discrete treatment literature. Consider the binary treatment setting where $D_i \in \{0, 1\}$. Let $e(X_i) = \mathbb{P}(D_i = 1 \mid X_i)$, $\sigma_1^2(X_i) = Var(Y_i \mid D_i = 1, X_i)$, and $\sigma_0^2(X_i) = Var(Y_i \mid D_i = 0, X_i)$. The precision-weighted propensity becomes

$$e_\omega(X_i) = \frac{e(X_i)/\sigma_1^2(X_i)}{e(X_i)/\sigma_1^2(X_i) + (1 - e(X_i))/\sigma_0^2(X_i)}.$$

The target population weight implied by the optimal continuous balancing weight for treated units is $w^*(X_i) = \alpha_\omega^*(1, X_i)e(X_i)$. Simplification of Theorem 1(a) under this discrete support reveals

$$w^*(X_i) \propto \left(\frac{\sigma_1^2(X_i)}{e(X_i)} + \frac{\sigma_0^2(X_i)}{1 - e(X_i)} \right)^{-1}.$$

This equivalence recovers the optimal heteroskedastic overlap weights established in Crump et al. (2006).

Second, when the noise varies across covariates but not across doses, the optimal estimand is analogous to Feasible Generalized Least Squares (FGLS). It inversely weights subpopulations by their outcome variance $\sigma^2(X_i)$, giving more weights to covariate strata with more precisely measured outcomes.

Third, under homoskedasticity, the optimal continuous estimand τ^* is algebraically equivalent to the population projection parameter from a Partially Linear Regression (PLR) (Robinson, 1988). In empirical practice, the PLR coefficient is often justified as a causal parameter only under the restrictive assumption of constant treatment effects. This equivalence establishes an efficiency-based causal interpretation for the PLR parameter even when treatment effects are heterogeneous. It identifies the weighted average derivative effect that minimizes the nonparametric efficiency bound. Furthermore, adapting the optimization criterion under homoskedasticity to minimize the sum of the asymptotic variances for all pairwise comparisons recovers the Generalized Overlap Weights (Li and Li, 2019) (See Appendix B).

Remark 2. All of our 3 optimal estimands satisfy the non-negativity property by Lemma 1. From Theorem 1, $\alpha_\omega^*(d, x)$ is proportional to $\omega(d, x)(d - e_\omega(x))$ scaled by a positive denominator. We evaluate the sign of the unscaled truncated expectation $I(d, x) = \mathbb{E}[\omega(D_i, X_i)(D_i - e_\omega(x))\mathbb{I}(D_i \le d) \mid X_i = x]$ in two regions. For $d \le e_\omega(x)$, the term $(D_i - e_\omega(x))$ is non-positive almost surely on the event $\{D_i \le d\}$. Since the precision weight is strictly positive

$\omega(D_i, X_i) > 0$, it follows that $I(d, x) \leq 0$. For $d > e_\omega(x)$, we utilize the covariate balance constraint. By the definition of $e_\omega(x)$, the full conditional expectation is zero, meaning $\mathbb{E}[\omega(D_i, X_i)(D_i - e_\omega(x)) \mid X_i = x] = 0$. We can therefore rewrite the truncated expectation as the negation of the upper tail $I(d, x) = -\mathbb{E}[\omega(D_i, X_i)(D_i - e_\omega(x))\mathbb{I}(D_i > d) \mid X_i = x]$. On the event $\{D_i > d\}$, we have $D_i > d > e_\omega(x)$, making the term inside the expectation positive. Its negation is therefore negative, yielding $I(d, x) < 0$. Therefore, $\mathbb{E}[\alpha^*_\omega(D_i, X_i) \mid D_i \leq d, X_i = x] \leq 0$ for all d and x.

3.3 The Efficient Influence Function

Theorem 1 characterizes the optimal continuous estimands assuming the structural components, such as ω, e_ω, and μ, are known. In practice, these non-parametric nuisance parameters must be estimated. To estimate the optimal estimands at the parametric $\sqrt{n}$-rate using flexible machine learning methods, modern semiparametric theory requires us to find the Efficient Influence Function (EIF) of the target parameter. The EIF automatically provides the Neyman orthogonal score necessary to immunize the estimator against the slow convergence rates of the nuisance parameters (Chernozhukov et al., 2018).

To analyze the EIF comprehensively across all three heteroskedastic architectures, we construct a unified projection representation. We define the precision-weighted conditional outcome mean $\rho_\omega(x)$ as

$$\rho_\omega(x) = \frac{\mathbb{E}[\omega(D_i, X_i)\mu(D_i, X_i) \mid X_i = x]}{\mathbb{E}[\omega(D_i, X_i) \mid X_i = x]}. \tag{10}$$

Applying the law of iterated expectations, all three optimal estimands τ^*_ω defined in Theorem 1 can be written as a projection

$$\tau^*_\omega = \frac{\mathbb{E}\left[\omega(D_i, X_i)\big(D_i - e_\omega(X_i)\big)\big(\mu(D_i, X_i) - \rho_\omega(X_i)\big)\right]}{\mathbb{E}\left[\omega(D_i, X_i)\big(D_i - e_\omega(X_i)\big)^2\right]}. \tag{11}$$

We now derive the EIF for τ^*_ω. Let $\eta = (\omega, \mu, e_\omega, \rho_\omega)$ denote the vector of nuisance parameters. Because the precision weight $\omega = \sigma^{-2}$ depends on the conditional variance of the outcome, estimating it introduces a specific adjustment into the influence function. To capture this analytically, we define the debiased precision weight as

$$\omega_{db}(O_i;\eta) = 2\omega(D_i,X_i) - \omega^2(D_i,X_i)\big(Y_i - \mu(D_i,X_i)\big)^2. \tag{12}$$

Notice that when evaluated at the true parameters η_0, the conditional expectation satisfies $\mathbb{E}[\omega_{db}(O_i;\eta_0) \mid D_i, X_i] = 2\omega_0 - \omega_0^2\sigma_0^2 = \omega_0$. The conditional expectation of the debiased weight exactly recovers the true precision weight, but its tracking of the squared outcome residuals ensures the Neyman orthogonality.

Theorem 2 (Efficient Influence Function of Optimal Estimands)**.** *Suppose Assumption 2 holds, and let η_0 denote the true nuisance parameters. The Efficient Influence Function (EIF) of τ^*_ω is given by $\phi^*(O_i) = K_{\omega_0}^{-1}\psi_{aug}(O_i;\tau^*_\omega,\eta_0)$, where $K_{\omega_0} = \mathbb{E}[\omega_0(D_i,X_i)(D_i - e_{\omega,0}(X_i))^2]$, and the unscaled augmented score is $\psi_{aug}(O_i;\tau,\eta) = A_{aug}(O_i;\eta) - \tau B_{aug}(O_i;\eta)$, with components*

$$\begin{aligned} A_{aug}(O_i;\eta) &= \omega_{db}(O_i;\eta)\big(D_i - e_\omega(X_i)\big)\big(\mu(D_i,X_i) - \rho_\omega(X_i)\big) \\ &\qquad + \omega(D_i,X_i)\big(D_i - e_\omega(X_i)\big)\big(Y_i - \mu(D_i,X_i)\big), \end{aligned} \tag{13}$$

$$B_{aug}(O_i;\eta) = \omega_{db}(O_i;\eta)\big(D_i - e_\omega(X_i)\big)^2. \tag{14}$$

Theorem 2 characterizes the statistical limit for estimating continuous treatment effects and highlights three theoretical properties. First, the EIF natively bypasses the ill-posed density derivative estimation ($\partial_d \log f(d|x)$). Second, the score is insensitive to the estimation of nuisance parameters e_ω and ρ_ω because the estimand is defined via an orthogonal projection. Finally, the appearance of the debiased weight ω_{db} corrects for the first-order bias induced by estimating the structural precision $\omega(d,x)$.

4 Estimation of the Optimal Estimands

The optimal continuous balancing weights derived in Theorem 1 identify a class of target estimands τ^*_ω that minimize the nonparametric efficiency bound under various heteroskedastic scenarios. In practice, evaluating these estimands requires estimating the nonparametric nuisance functions $\eta = (\omega,\mu,e_\omega,\rho_\omega)$. Replacing these unknown functions with flexible machine learning estimates in standard plug-in estimators generally fails to achieve parametric $\sqrt{n}$-consistency due to regularization bias and slow convergence rates. To overcome this, we leverage the EIF derived in Theorem 2 and develop $\sqrt{n}$-consistent estimators based on the DML framework.

The estimation procedure is described as follows

1. Randomly partition the sample indices $\{1, \ldots, n\}$ into K folds, $\mathcal{I}_1, \ldots, \mathcal{I}_K$. Let $\mathcal{I}_k^c$ denote the out-of-fold complement for fold k.

2. For each fold $k \in \{1, \ldots, K\}$, estimate the nuisance parameters.

   2a. Train machine learning models to estimate the conditional mean $\hat{\mu}_k(d, x)$ by regressing Y_i on (D_i, X_i). Then, estimate the conditional variance $\hat{\sigma}_k^2(d, x)$ by regressing the squared residuals $(Y_i - \hat{\mu}_k(D_i, X_i))^2$ on (D_i, X_i). Construct the precision weight as $\hat{\omega}_k(d, x) = 1/\max\{\hat{\sigma}_k^2(d, x), c\}$ for some small trimming constant $c > 0$.

   2b. Estimate $\hat{e}_{\omega,k}(x)$ and $\hat{\rho}_{\omega,k}(x)$ by passing $\hat{\omega}_k(D_i, X_i)$ to the ML models trained on $\mathcal{I}_k^c$

$$\hat{e}_{\omega,k} = \arg\min_{f \in \mathcal{F}} \sum_{j \in \mathcal{I}_k^c} \hat{\omega}_k(D_j, X_j)\big(D_j - f(X_j)\big)^2,$$

$$\hat{\rho}_{\omega,k} = \arg\min_{g \in \mathcal{G}} \sum_{j \in \mathcal{I}_k^c} \hat{\omega}_k(D_j, X_j)\big(Y_j - g(X_j)\big)^2.$$

3. Construct the final cross-fitted estimator $\hat{\tau}_\omega^*$ by solving the empirical moment condition $\frac{1}{n}\sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \psi_{aug}(O_i; \hat{\tau}_\omega^*, \hat{\eta}_k) = 0$, which yields

$$\hat{\tau}_\omega^* = \frac{\sum_{k=1}^K \sum_{i \in \mathcal{I}_k} A_{aug}(O_i; \hat{\eta}_k)}{\sum_{k=1}^K \sum_{i \in \mathcal{I}_k} B_{aug}(O_i; \hat{\eta}_k)}. \tag{15}$$

Theorem 3 (Asymptotic Normality)**.** *Suppose the conditions in Assumption 2 hold, and let $\eta_0 = (\omega_0, \mu_0, e_{\omega,0}, \rho_{\omega,0})$ denote the true nuisance parameters, and let $\hat{\eta}_k = (\hat{\omega}_k, \hat{\mu}_k, \hat{e}_{\omega,k}, \hat{\rho}_{\omega,k})$ denote their cross-fitted machine learning estimators. Assume that at least one of the following two conditions holds:*

- *Condition A:*

  (i) There exists $C > 0$ such that $|Y_i| \le C$ and $|D_i| \le C$ almost surely.

(ii) $\|\hat{\mu}_k\|_\infty$, $\|\hat{e}_{\omega,k}\|_\infty$, $\|\hat{\rho}_{\omega,k}\|_\infty \leq C$.

(iii) $\|\hat{\mu}_k - \mu_0\|_{P,2}$, $\|\hat{\omega}_k - \omega_0\|_{P,2}$, $\|\hat{e}_{\omega,k} - e_{\omega,0}\|_{P,2}$, *and* $\|\hat{\rho}_{\omega,k} - \rho_{\omega,0}\|_{P,2}$ *are all* $o_P(n^{-1/4})$.

- *Condition B:*

  (i) There exists $q \geq 8$ *such that* $\mathbb{E}[|Y_i|^q] < \infty$ *and* $\mathbb{E}[|D_i|^q] < \infty$.

  (ii) $\|\hat{\mu}_k - \mu_0\|_\infty$, $\|\hat{\omega}_k - \omega_0\|_\infty$, $\|\hat{e}_{\omega,k} - e_{\omega,0}\|_\infty$, *and* $\|\hat{\rho}_{\omega,k} - \rho_{\omega,0}\|_\infty$ *are all* $o_P(1)$.

  (iii) Let $v(O_i) = 1 + |D_i - e_{\omega,0}(X_i)| + |\mu_0(D_i, X_i) - \rho_{\omega,0}(X_i)|$. $\|(\hat{\mu}_k - \mu_0)v(O)\|_{P,2}$, $\|(\hat{\omega}_k - \omega_0)v(O)\|_{P,2}$, $\|\hat{e}_{\omega,k} - e_{\omega,0}\|_{P,2}$, *and* $\|\hat{\rho}_{\omega,k} - \rho_{\omega,0}\|_{P,2}$ *are all* $o_P(n^{-1/4})$.

Then, the estimator in Equation (15) is a Regular Asymptotically Linear (RAL) estimator of τ^*_ω, *i.e.,*

$$\sqrt{n}(\hat{\tau}^*_\omega - \tau^*_\omega) \xrightarrow{d} \mathcal{N}(0, V^*_\omega),$$

where

$$V^*_\omega = \frac{\mathbb{E}[\psi^2_{aug}(O_i; \tau^*_\omega, \eta_0)]}{(\mathbb{E}[\omega_0(D_i, X_i)(D_i - e_{\omega,0}(X_i))^2])^2},$$

which can be consistently estimated by plugging in the sample analogues.

Remark 3. The convergence requirements in Theorem 3 are stronger than those for the baseline DML (Chernozhukov et al., 2018). Because estimating the heteroskedastic optimal estimand requires the precision weight $\omega(d, x)$, and debiasing the precision weight must track squared outcome residuals. This introduces higher-order polynomial terms into the Neyman orthogonal remainder (e.g., $(\hat{\mu} - \mu_0)^2(\hat{e} - e_0)^2$), for which the standard Cauchy-Schwarz bounds does not apply. Theorem 3 provides two sets of conditions to handle the higher-order terms. For variables with bounded support (Condition A), the complex remainders are uniformly bounded, relaxing the requirement back to the standard $L_2(P)$ rate of $o_P(n^{-1/4})$. Condition A(ii) can be satisfied by trimming the estimators or by utilizing natively bounded ML estimators such as tree-based ensembles. For unbounded variables (Condition B), we control the remainders by requiring uniform consistency ($\|\hat{\eta}_k - \eta_0\|_\infty = o_P(1)$) alongside $L_2(P)$ rate of $o_P(n^{-1/4})$ for nuisance estimators weighted by an "envelope function" $v(O_i)$. Condition B(i) guarantees that the envelope $v(O_i)$ possesses finite 4th moments, so its norm $\|v(O)\|_{P,4}$ is a bounded constant. Consequently, this requirement reduces to the underlying nuisance estimators achieving the $o_P(n^{-1/4})$ convergence rate.

Remark 4. We Note that the conditions in Theorem 3 are required for the estimation and inference under general heteroskedasticity. However, the conditions required under homoskedasticity are remarkably relaxed. When $\omega = 1$, $\mu(d, x)$ cancels from the augmented score, reducing to the classic Partially Linear Regression (PLR) score

$$\psi_{homo}(O_i; \tau, \eta) = \big(D_i - e(X_i)\big)\big(Y_i - \rho(X_i)\big) - \tau\big(D_i - e(X_i)\big)^2. \tag{16}$$

The estimation of PLR has been intensively studied in the literature (e.g., Robinson, 1988; Chernozhukov et al., 2018). Because its remainder is bilinear, $\sqrt{n}$-consistency requires only standard $o_P(n^{-1/4})$ rate for nuisance estimators, recovering Theorem 3.1 of Chernozhukov et al. (2018). Crucially, our continuous balancing framework formalizes a causal justification for deploying the PLR estimator even when the true treatment effect is heterogeneous. Theorem 1(c) and Proposition 2 prove that PLR coefficient identifies a well-defined, variance-weighted average effect that prioritizes subpopulations with high $Var(D_i \mid X_i)$. This estimand also guarantees non-negative weights via Lemma 1. Thus, researchers can safely trade the absolute asymptotic efficiency of the general estimand for relaxed statistical requirements and finite-sample robustness by using PLR.

5 Simulation Study

In this section, we conduct a Monte Carlo simulation study to evaluate the finite-sample performance of the proposed continuous balancing estimators. We draw a four-dimensional pre-treatment covariate vector, $X_i = (X_{i1}, X_{i2}, X_{i3}, X_{i4})'$, from a standard uniform distribution, $\mathcal{U}[0, 1]^4$. Note that X_{i4} is a noise variable that does not enter the data generating process.

The data generating process is

$$\begin{aligned}
D_i|X_i &\sim \mathcal{B}\text{eta}\big(2 + X_{i1}, 2 + X_{i2}\big),\\
\mu(D_i, X_i) &= D_i - 0.5D_i^2 + X_{i1}D_i + X_{i2} + X_{i3}^2,\\
Y_i &= \mu(D_i, X_i) + \sigma(D_i, X_i)\varepsilon_i, \quad \varepsilon_i \sim \mathcal{TN}_{[-3,3]}(0, 1),
\end{aligned}$$

where the idiosyncratic error ε_i is drawn from a truncated normal distribution on the interval $[-3, 3]$. The true marginal derivative effect is heterogeneous across both dosage and covariates, given by $\mu'(D_i, X_i) = 1 - D_i + X_{i1}$. To evaluate the efficiency bounds derived in

Theorem 1, we create three heteroskedastic scenarios

$$\begin{aligned} \text{Homoskedastic:} \quad & \sigma(D_i, X_i) = 0.5 \\ \text{Covariate-Dependent:} \quad & \sigma(D_i, X_i) = 0.5\sqrt{1 + 2X_{i1}^2} \\ \text{General Heteroskedasticity:} \quad & \sigma(D_i, X_i) = 0.5\sqrt{1 + 2X_{i1}^2 + 4D_i^2} \end{aligned}$$

Because the marginal treatment effect is heterogeneous, the theoretically optimal estimands identify distinct population parameters. Bias of each estimator is then calculated relative to their corresponding estimand.

We compare four estimators. The first three are our proposed optimal balancing estimators: the homoskedastic PLR estimand $\hat{\tau}^*_{homo}$, the covariate-dependent optimal estimand $\hat{\tau}^*_{cov}$, and the general optimal estimand $\hat{\tau}^*_{gen}$. These are implemented via the estimation procedure in Section 4 with 5-fold cross-fitting and trimming constant $c = 0.01$, employing extreme gradient boosting (xgboost) for all nuisance functions.

Estimator 4 is the baseline Average Derivative Estimator (ADE) implemented via the Generalized Propensity Score (GPS) method by Hirano and Imbens (2004). We estimate the GPS by using a series estimator to obtain the conditional density $f(d|x)$, and then estimate the outcome's conditional expectation via a quadratic polynomial regression. We average this estimated dose-response function over the empirical distribution of X_i, and take its analytical derivative to construct $\hat{\tau}_{ADE}$. The standard error is calculated using the bootstrap method with 1,000 bootstrap samples.

We conduct 1,000 Monte Carlo simulations for sample sizes $n = 500$ and $n = 2000$. Table 1 reports the mean estimate, standard error, bias, and coverage probability of the 95% confidence intervals for each estimator. Across all designs, the baseline GPS estimator exhibits large finite-sample bias, inflated standard errors, and severe under-coverage, compared to our proposed estimators. Each proposed estimator achieves the lowest standard error when its precision-weighting scheme matches the true heteroskedastic scenario of the DGP.

Examining the moderate sample size reveals a critical finite-sample trade-off. Under general heteroskedasticity, the theoretically optimal $\hat{\tau}^*_{gen}$ yields a higher standard error than $\hat{\tau}^*_{homo}$. Achieving full efficiency requires nonparametrically estimating the variance function $\sigma^2(d, x)$.

With limited data, the estimation of this nuisance function introduces finite-sample noise into the weights. This observation reflects our discussion of the robustness and convenience of the PLR estimand in Section 4.

6 Application: The Income Effect on Labor Supply

We illustrate our proposed optimal estimands by re-examining the effect of unearned income on labor supply, utilizing the dataset from a survey of Massachusetts lottery winners analyzed by Imbens et al. (2001) and Hirano and Imbens (2004) (hereafter HI04). The objective is to estimate the Marginal Propensity to Earn (MPE) out of unearned income, which corresponds to the average derivative of labor earnings with respect to the lottery prize.

The dataset consists of 237 individuals who won the Megabucks lottery in the mid-1980s. The continuous treatment D_i is the annualized value of the lottery prize, and the outcome Y_i is the average labor earnings recorded by the Social Security Administration approximately six years after winning. The covariates X_i include demographic characteristics (age, gender, education) and six years of pre-lottery earnings.

While the lottery prize itself is randomly assigned, the study design introduced potential confounding due to substantial nonresponse (around 50%). HI04 demonstrated that non-response was correlated with the prize amount: winners of larger prizes were less likely to participate in the survey. This induced selection bias in the observed sample. For instance, men and individuals with higher pre-lottery earnings tended to have won larger prizes among the respondents. Following the literature, we maintain the assumption that conditional on the extensive set of covariates X_i, the treatment assignment is unconfounded.

HI04 addressed this setting by introducing the Generalized Propensity Score (GPS) methodology to estimate the entire dose-response curve, $\mu(d) = \mathbb{E}[Y_i(d)]$. Their analysis revealed significant heterogeneity in the MPE, $\mu'(d)$. They estimated that the MPE ranged from approximately -0.10 for small prizes (\$10,000) to -0.02 for larger prizes (\$100,000), suggesting the negative income effect is substantially stronger at lower levels of unearned income.

We compare four estimators from Section 5 for summarizing the treatment effect. [1] Table 2

[1] We use the data provided in the GitHub repository of Imbens and Xu (2024). We failed to reproduce

Table 1: Monte Carlo Simulation Results

Design	Estimator	Estimate	Std. Error	Bias	95% Cov.
Panel A: Moderate Sample ($n = 500$)					
1. Homoskedastic	$\hat{\tau}^*_{homo}$	0.984	0.045	-0.002	0.939
	$\hat{\tau}^*_{cov}$	0.897	0.051	-0.004	0.921
	$\hat{\tau}^*_{gen}$	0.969	0.056	0.004	0.920
	$\hat{\tau}_{ADE}$	0.865	0.125	-0.135	0.785
2. Covariate-Dep.	$\hat{\tau}^*_{homo}$	0.981	0.065	-0.005	0.937
	$\hat{\tau}^*_{cov}$	0.903	0.058	0.002	0.925
	$\hat{\tau}^*_{gen}$	0.961	0.063	-0.004	0.913
	$\hat{\tau}_{ADE}$	0.840	0.140	-0.160	0.740
3. General Het.	$\hat{\tau}^*_{homo}$	0.980	0.071	-0.006	0.932
	$\hat{\tau}^*_{cov}$	0.896	0.076	-0.005	0.918
	$\hat{\tau}^*_{gen}$	0.973	0.082	0.008	0.914
	$\hat{\tau}_{ADE}$	0.815	0.165	-0.185	0.680
Panel B: Large Sample ($n = 2000$)					
1. Homoskedastic	$\hat{\tau}^*_{homo}$	0.985	0.022	-0.001	0.950
	$\hat{\tau}^*_{cov}$	0.902	0.024	0.001	0.948
	$\hat{\tau}^*_{gen}$	0.967	0.026	0.002	0.945
	$\hat{\tau}_{ADE}$	0.895	0.065	-0.105	0.815
2. Covariate-Dep.	$\hat{\tau}^*_{homo}$	0.984	0.033	-0.002	0.949
	$\hat{\tau}^*_{cov}$	0.902	0.028	0.001	0.952
	$\hat{\tau}^*_{gen}$	0.967	0.030	0.002	0.948
	$\hat{\tau}_{ADE}$	0.875	0.072	-0.125	0.760
3. General Het.	$\hat{\tau}^*_{homo}$	0.983	0.037	-0.003	0.948
	$\hat{\tau}^*_{cov}$	0.899	0.036	-0.002	0.945
	$\hat{\tau}^*_{gen}$	0.966	0.031	0.001	0.951
	$\hat{\tau}_{ADE}$	0.845	0.088	-0.155	0.705

Note: Because the treatment effect is heterogeneous, shifting the precision-weighting scheme alters the targeted population parameter. Therefore, bias is calculated relative to each estimator's corresponding estimand.

presents the estimation results. All estimates confirm a negative income effect, although the magnitudes vary slightly, reflecting the distinct estimands implied by the different weighting schemes.

Table 2: Estimates of the Marginal Propensity to Earn (MPE) out of Lottery Prizes

Estimand	Method	Estimate	Std. Error	95% CI
ADE (τ_{ADE})	GPS	-0.0308	(0.0270)	[-0.1069, -0.0001]
Homoskedastic (τ^*_{homo})	DML	-0.0402	(0.0125)	[-0.0648, -0.0156]
Covariate-Dep. (τ^*_{cov})	DML	-0.0279	(0.0118)	[-0.0512, -0.0046]
General Het. (τ^*_{gen})	DML	-0.0253	(0.0107)	[-0.0463, -0.0044]

Proposition 2 and Theorem 1 can be used to provide a transparent interpretation of the target population. For example, τ^*_{homo} is a weighted average derivative where the weights are proportional to the conditional variance of the treatment, $Var(D_i|X_i)$. It prioritizes individuals for whom the prize amount is most variable given their covariates. To interpret this weighting in the context of the lottery data, we consider the structure of $Var(D_i|X_i)$. HI04 modeled the log of the prize as homoskedastic $\log(D_i)|X_i \sim \mathcal{N}(\beta' X_i, \sigma^2)$. Under this log-normal structure, the conditional variance of the prize in levels is heteroskedastic and proportional to $\exp(2\beta' X_i)$. This implies the variance is higher for individuals whose characteristics are associated with larger expected prizes.

Since HI04 documented that men and individuals with higher pre-lottery earnings tended to have won larger prizes in the sample, due to the nonresponse bias, τ^*_{homo} structurally places greater weight on these groups. Our estimate therefore reflects the income effect for this specific demographic subpopulation. This application demonstrates how our framework provides an efficiency-based causal estimand, clarifying the exact population for which the causal effect is being summarized.

7 Conclusion

Estimating the causal effects of continuous treatments requires summarizing the dose-response curve in a way that is both interpretable and statistically stable. This endeavor faces two

the exact results of HI04, as the summary statistics of pre-lottery earnings are slightly different from those reported in HI04. Our replication results are provided in Appendix C.

challenges. First, traditional summaries, such as the average derivative effect, are often statistically unstable because their estimation relies on the non-smooth conditional density function. Second, while the literature offered a path to optimality in categorical settings, generalizing it to continuous treatments presented a significant theoretical hurdle. Standard efficiency theories rely on the smoothness of derivative weights, yet this condition often fails in continuous settings.

This paper overcomes this barrier by extending the balancing weights framework to the continuous treatment setting. Rather than specifying a derivative weight and deriving an unstable estimand, we directly define the target estimand by specifying an outcome weight. This structural shift ensures the statistical functional is bounded by design, completely bypassing the need for density derivative estimation and providing a unified foundation for continuous efficiency analysis.

Our main contribution is the characterization of optimal continuous balancing weights that minimize the nonparametric efficiency bound. We derive a class of optimally efficient estimands under general heteroskedastic, covariate-dependent, and homoskedastic scenarios. We then derive their Efficient Influence Functions, and establish $\sqrt{n}$-consistent Double Machine Learning estimators.

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A Appendix: Proofs

Note: For notational simplicity, we omit individual subscripts in this appendix, writing X instead of X_i, Y instead of Y_i, D instead of D_i, etc.

Assumption 3 (Regularity Condition for Powell et al. (1989))**.** *a. The weighted conditional density $w(d,x)f(d|x)$ is differentiable with respect to d.*

b. $w(\underline{d},x)f(\underline{d}|x) = w(\bar{d},x)f(\bar{d}|x) = 0$ where $\underline{d}$ and $\bar{d}$ are the lower and upper bounds of the support of D.

c. $f(d|x) = 0$ implies $w(d,x) = 0$.

Proof of Proposition 1. We evaluate the causal target parameter by analyzing the limit of the expected incremental effect

$$\dot{\tau}_w = \lim_{\nu\to 0} \mathbb{E}\left[w(D_i,X_i)\frac{Y_i(D_i+\nu)-Y_i(D_i)}{\nu}\right].$$

By the law of iterated expectations, conditioning on the observed treatment and covariates yields

$$\dot{\tau}_w = \lim_{\nu\to 0} \mathbb{E}\left[w(D_i,X_i)\mathbb{E}\left[\frac{Y_i(D_i+\nu)-Y_i(D_i)}{\nu} \;\middle|\; D_i,X_i\right]\right].$$

By Assumption 1(a) and 1(b), the conditional expectation of the potential outcome evaluated at any generic level d satisfies $\mathbb{E}[Y_i(d)|D_i,X_i] = \mathbb{E}[Y_i(d)|X_i] = m(d,X_i)$. Applying this equality at $d = D_i+\nu$ and $d = D_i$, the inner conditional expectation simplifies to

$$\mathbb{E}\left[\frac{Y_i(D_i+\nu)-Y_i(D_i)}{\nu} \;\middle|\; D_i,X_i\right] = \frac{m(D_i+\nu,X_i)-m(D_i,X_i)}{\nu}.$$

Thus, the estimand becomes

$$\dot{\tau}_w = \lim_{\nu\to 0} \mathbb{E}\left[w(D_i,X_i)\frac{m(D_i+\nu,X_i)-m(D_i,X_i)}{\nu}\right].$$

By the Mean Value Theorem, there exists some intermediate value D^* between D_i and $D_i+\nu$ such that the difference quotient equals $\partial_d m(D^*,X_i)$. Under Assumption 1(d), the derivative $\partial_d m(d,x)$ is uniformly bounded by a constant $C > 0$. Thus, the absolute value of the argument inside the expectation is bounded by $C|w(D_i,X_i)|$. Since $\mathbb{E}[|w(D_i,X_i)|] < \infty$

by Assumption 1(e), the random variable is dominated by an integrable function. The Dominated Convergence Theorem applies, allowing us to pass the limit inside the expectation

$$\begin{aligned}\dot{\tau}_w &= \mathbb{E}\left[w(D_i, X_i) \lim_{\nu \to 0} \frac{m(D_i + \nu, X_i) - m(D_i, X_i)}{\nu}\right] \\ &= \mathbb{E}\left[w(D_i, X_i) \partial_d m(D_i, X_i)\right].\end{aligned}$$

Finally, we connect the structural derivative $\partial_d m$ to the observable regression derivative μ'. For any observed point (d_0, x_0) in the support of P, Assumption 1(c) guarantees $f(d|x_0) > 0$ within an open neighborhood $\mathcal{N}(d_0)$. For any $d \in \mathcal{N}(d_0)$, the observable regression function satisfies

$$\mu(d, x_0) = \mathbb{E}[Y_i | D_i = d, X_i = x_0] = \mathbb{E}[Y_i(d) | D_i = d, X_i = x_0] = \mathbb{E}[Y_i(d) | X_i = x_0] = m(d, x_0).$$

Because $\mu(d, x)$ is identical to $m(d, x)$ over an open neighborhood, their partial derivatives with respect to d must be equal everywhere in the support: $\mu'(D_i, X_i) = \partial_d m(D_i, X_i)$. Substituting this equality recovers the statistical functional $\dot{\tau}_w = \mathbb{E}[w(D_i, X_i)\mu'(D_i, X_i)]$. □

Proof of Proposition 2. We show two results. First, the estimand defined by $\alpha \in \mathcal{A}$ corresponds to a weighted average derivative with the specified weight $w(d, x)$. Second, this weight is normalized, i.e., $\mathbb{E}[w(D, X)] = 1$.

Relating α and w

We need to verify that for any sufficiently smooth and bounded function $\mu \in \mathcal{H}$, the following identity holds

$$\mathbb{E}[\alpha(D, X)\mu(D, X)] = \mathbb{E}[w(D, X)\mu'(D, X)]. \tag{A.1}$$

We analyze this identity conditionally on $X = x$. Let $w(d, x)$ be defined as in Equation (2)

$$w(d, x) = -\frac{F(d|x)}{f(d|x)} \cdot \mathbb{E}[\alpha(D, X) | D \leq d, X = x].$$

Let $H(d, x) = w(d, x) f(d|x)$. We can rewrite $H(d, x)$ using the definition of conditional expectation

$$
\begin{aligned}
H(d,x) &= -F(d|x)\cdot \mathbb{E}[\alpha(D,X)|D\le d, X=x] \\
&= -\mathbb{E}[\alpha(D,X)\mathbb{I}(D\le d)|X=x] \\
&= -\int_{\underline{d}}^{d} \alpha(t,x)f(t|x)\mathrm{d}t.
\end{aligned}
$$

We evaluate the right-hand side of Equation (A.1) conditional on $X=x$ using integration by parts

$$
\begin{aligned}
\mathbb{E}[w(D,X)\mu'(D,X)|X=x] &= \int_{\underline{d}}^{\bar{d}} w(d,x)\mu'(d,x)f(d|x)\mathrm{d}d \\
&= \int_{\underline{d}}^{\bar{d}} \mu'(d,x)H(d,x)\mathrm{d}d \\
&= [\mu(d,x)H(d,x)]_{\underline{d}}^{\bar{d}} - \int_{\underline{d}}^{\bar{d}} \mu(d,x)\frac{\partial H(d,x)}{\partial d}\mathrm{d}d.
\end{aligned}
$$

We must verify that the boundary terms vanish. At the lower bound $\underline{d}$

$$
H(\underline{d},x) = -\int_{\underline{d}}^{\underline{d}} \alpha(t,x)f(t|x)\mathrm{d}t = 0.
$$

At the upper bound $\bar{d}$

$$
H(\bar{d},x) = -\int_{\underline{d}}^{\bar{d}} \alpha(t,x)f(t|x)\mathrm{d}t = -\mathbb{E}[\alpha(D,X)|X=x].
$$

Since α is a continuous balancing weight with $\mathbb{E}[\alpha(D,X)|X=x]=0$. Thus, $H(\bar{d},x)=0$. The boundary terms vanish (since μ is bounded by Assumption 1d).[2]

Now we analyze the derivative of $H(d,x)$. By the Fundamental Theorem of Calculus

$$
\frac{\partial H(d,x)}{\partial d} = \frac{\partial}{\partial d}\left(-\int_{\underline{d}}^{d} \alpha(t,x)f(t|x)\mathrm{d}t\right) = -\alpha(d,x)f(d|x).
$$

[2]If the support is unbounded (e.g., $\mathbb{R}$), we assume standard regularity conditions such that $\lim_{d\to\pm\infty}\mu(d,x)H(d,x)=0$.

Substituting this back into the integration by parts formula

$$\begin{aligned}\mathbb{E}[w(D,X)\mu'(D,X)|X=x] &= 0 - \int_{\underline{d}}^{\bar{d}} \mu(d,x)[-\alpha(d,x)f(d|x)]\mathrm{d}d \\ &= \int_{\underline{d}}^{\bar{d}} \mu(d,x)\alpha(d,x)f(d|x)\mathrm{d}d \\ &= \mathbb{E}[\alpha(D,X)\mu(D,X)|X=x].\end{aligned}$$

This confirms the identity in Equation (A.1).

Normalization of $w(d,x)$

We must show that $\mathbb{E}[w(D,X)] = 1$. We utilize the identity established in Part 1, which holds for any smooth function $\mu(d,x)$. We choose the test function $\mu(d,x) = d$. Then $\mu'(d,x) = 1$.

Substituting this into the identity $\mathbb{E}[w\mu'|X] = \mathbb{E}[\alpha\mu|X]$

$$\mathbb{E}[w(D,X)\cdot 1|X] = \mathbb{E}[\alpha(D,X)\cdot D|X].$$

Taking the expectation over X using the Law of Iterated Expectations

$$\begin{aligned}\mathbb{E}[w(D,X)] &= \mathbb{E}[\mathbb{E}[w(D,X)|X]] \\ &= \mathbb{E}[\mathbb{E}[\alpha(D,X)D|X]] \\ &= \mathbb{E}[\alpha(D,X)D].\end{aligned}$$

Since α is a continuous balancing weight, the normalization constraint ensures $\mathbb{E}[\alpha(D,X)D] = 1$. Therefore, $\mathbb{E}[w(D,X)] = 1$. □

Proof of Lemma 1. From Equation (2), since $F(d|x) \geq 0$ and $f(d|x) \geq 0$ for all d, x, it is straightforward to show that $w(d,x) \geq 0$ if and only if

$$\mathbb{E}[\alpha(D,X)|D \leq d, X = x] \leq 0 \text{ for all } d, x.$$

□

Proof of Theorem 1. We seek the optimal continuous balancing weight α_ω that minimizes the variance component $V_S = \mathbb{E}[\alpha_\omega^2(D_i, X_i)\sigma^2(D_i, X_i)]$. Let $\omega(D_i, X_i) = \sigma^{-2}(D_i, X_i)$ denote the true precision weight. The objective simplifies to minimizing $\mathbb{E}[\alpha_\omega^2(D_i, X_i)\omega^{-1}(D_i, X_i)]$. Let $\mathcal{H} = L_2(P_{D,X})$. Under Assumption 2, $\omega(D_i, X_i)$ is strictly positive and bounded, ensuring the objective is well-defined. We formulate this as a constrained functional optimization problem in the Hilbert space $\mathcal{H}$,

$$\begin{aligned}
\min_{\alpha \in \mathcal{H}} \quad & \mathbb{E}\big[\alpha^2(D, X)\omega^{-1}(D, X)\big] \\
\text{subject to} \quad & (C1) : \mathbb{E}[\alpha(D, X)D] = 1, \\
& (C2) : \mathbb{E}[\alpha(D, X) \mid X] = 0 \text{ almost surely.}
\end{aligned}$$

We employ the method of Lagrange multipliers. Let $\lambda \in \mathbb{R}$ be the multiplier for constraint (C1), and let $\eta(X) \in L_2(P_X)$ be the functional multiplier for (C2). The Lagrangian is

$$L(\alpha, \lambda, \eta) = \mathbb{E}\big[\alpha^2(D, X)\omega^{-1}(D, X) - 2\lambda\alpha(D, X)D - 2\eta(X)\alpha(D, X)\big] + 2\lambda.$$

To find the optimum, we compute the Gâteaux derivative of L with respect to α in an arbitrary direction $h \in \mathcal{H}$ and set it to zero

$$\begin{aligned}
\nabla_\alpha L(\alpha; h) &= \lim_{\epsilon \to 0} \frac{L(\alpha + \epsilon h, \lambda, \eta) - L(\alpha, \lambda, \eta)}{\epsilon} \\
&= \frac{d}{d\epsilon}\mathbb{E}\big[(\alpha + \epsilon h)^2\omega^{-1}(D, X) - 2\lambda(\alpha + \epsilon h)D - 2\eta(X)(\alpha + \epsilon h)\big]\Big|_{\epsilon=0} \\
&= 2\mathbb{E}\big[h(D, X)\big(\alpha(D, X)\omega^{-1}(D, X) - \lambda D - \eta(X)\big)\big] = 0.
\end{aligned}$$

Because the derivative must equal zero for all arbitrary test functions $h \in \mathcal{H}$, the term inside the parenthesis must be zero almost surely. This establishes the first-order condition

$$\alpha(D, X) = \omega(D, X)\big(\lambda D + \eta(X)\big).$$

We apply the continuous covariate balance constraint (C2), $\mathbb{E}[\alpha(D, X) \mid X] = 0$, to solve for $\eta(X)$

$$\mathbb{E}\big[\omega(D, X)(\lambda D + \eta(X)) \mid X\big] = 0 \implies \lambda\mathbb{E}[\omega(D, X)D \mid X] + \eta(X)\mathbb{E}[\omega(D, X) \mid X] = 0.$$

Solving for $\eta(X)$ directly yields the precision-weighted centering term:

$$\eta(X) = -\lambda \frac{\mathbb{E}[\omega(D,X)D \mid X]}{\mathbb{E}[\omega(D,X) \mid X]} \equiv -\lambda e_\omega(X).$$

Substituting $\eta(X)$ back into the expression for $\alpha(D,X)$ provides the unnormalized optimal weight

$$\alpha(D,X) = \lambda\omega(D,X)\big(D - e_\omega(X)\big).$$

Finally, we apply the normalization constraint (C1), $\mathbb{E}[\alpha(D,X)D] = 1$, to uniquely identify the scalar multiplier λ

$$1 = \lambda\mathbb{E}\big[\omega(D,X)\big(D - e_\omega(X)\big)D\big].$$

We decompose D as $D = (D - e_\omega(X)) + e_\omega(X)$, separating the expectation

$$1 = \lambda\mathbb{E}\big[\omega(D,X)\big(D - e_\omega(X)\big)^2\big] + \lambda\mathbb{E}\big[\omega(D,X)\big(D - e_\omega(X)\big)e_\omega(X)\big].$$

By the Law of Iterated Expectations, the rightmost term evaluates exactly to zero because

$$\begin{aligned}
&\mathbb{E}\big[\omega(D,X)(D - e_\omega(X))e_\omega(X)\big] \\
=&\mathbb{E}_X\Big[e_\omega(X)\mathbb{E}[\omega(D,X)D - \omega(D,X)e_\omega(X) \mid X]\Big] \\
=&\mathbb{E}_X\Big[e_\omega(X)\big(\mathbb{E}[\omega(D,X)D \mid X] - e_\omega(X)\mathbb{E}[\omega(D,X) \mid X]\big)\Big] \\
=&0,
\end{aligned}$$

which follows from the definition of $e_\omega(X)$. This leaves the scalar multiplier

$$\lambda = \Big(\mathbb{E}\big[\omega(D,X)\big(D - e_\omega(X)\big)^2\big]\Big)^{-1}.$$

Substituting λ completes the derivation of the unified formula for $\alpha^*_\omega(d,x)$. The optimal estimand $\tau^*_\omega = \mathbb{E}[\alpha^*_\omega(D,X)Y]$ follows trivially. The specific forms in cases (a), (b), and (c) arise by defining $\omega(d,x)$ as $\sigma^{-2}(d,x)$, $\sigma^{-2}(x)$, and 1, respectively. □

Proof of Theorem 2. We use the method of pathwise differentiation. Let P_ϵ be a regular one-dimensional parametric submodel passing through the true distribution P_0 at $\epsilon = 0$, with score function $S(O) = \frac{\partial}{\partial\epsilon}\log p_\epsilon(O)\big|_{\epsilon=0}$. Assumption 2 ensures all required moments are finite.

The estimand is $\tau^*_\omega = N_\omega / K_\omega$, where the numerator is

$$N_\omega = \mathbb{E}[\omega(D, X)(D - e_\omega(X))(\mu(D, X) - \rho_\omega(X))]$$

and the denominator is

$$K_\omega = \mathbb{E}[\omega(D, X)(D - e_\omega(X))^2]$$

By the functional Delta method, the EIF of the ratio is $\phi^*(O) = \frac{1}{K_\omega}(\phi_N(O) - \tau^*_\omega \phi_K(O))$, where ϕ_N and ϕ_K are the EIFs of the numerator and denominator.

We calculate the pathwise derivative of the numerator N_ω evaluated at $\epsilon = 0$, denoted with a tilde

$$\begin{aligned}\tilde{N}_\omega = \mathbb{E}\big[&\tilde{\omega}(D - e_\omega)(\mu - \rho_\omega) - \omega\tilde{e}_\omega(\mu - \rho_\omega) - \omega(D - e_\omega)\tilde{\rho}_\omega + \omega(D - e_\omega)\tilde{\mu} \\ &+ \omega(D - e_\omega)(\mu - \rho_\omega)S(O)\big].\end{aligned} \tag{A.2}$$

Because e_ω and ρ_ω are defined via orthogonal projections, $\mathbb{E}[\omega(D - e_\omega) \mid X] = 0$ and $\mathbb{E}[\omega(\mu - \rho_\omega) \mid X] = 0$. By the law of iterated expectations, any cross-terms involving $\tilde{e}_\omega$ or $\tilde{\rho}_\omega$ vanish

$$\mathbb{E}[\omega\tilde{e}_\omega(\mu - \rho_\omega)] = \mathbb{E}_X\left[\tilde{e}_\omega(X)\mathbb{E}[\omega(\mu - \rho_\omega) \mid X]\right] = 0, \quad \text{and} \quad \mathbb{E}[\omega(D - e_\omega)\tilde{\rho}_\omega] = 0.$$

The pathwise derivative of the conditional mean is $\tilde{\mu}(D, X) = \mathbb{E}[(Y - \mu)S(O) \mid D, X]$. Thus,

$$\mathbb{E}[\omega(D - e_\omega)\tilde{\mu}] = \mathbb{E}\big[\omega(D - e_\omega)(Y - \mu)S(O)\big].$$

We now evaluate the pathwise derivative of the precision weight $\tilde{\omega}$. Since $\omega(D, X) = 1/\sigma^2(D, X)$ where $\sigma^2(D, X) = \mathbb{E}[(Y - \mu)^2 \mid D, X]$, the chain rule yields:

$$\tilde{\omega} = -\frac{1}{\sigma^4}\tilde{\sigma}^2 = -\omega^2\mathbb{E}\big[((Y - \mu)^2 - \sigma^2)S(O) \mid D, X\big].$$

Substituting this into the first term of Equation (A.2):

$$\mathbb{E}\big[\tilde{\omega}(D - e_\omega)(\mu - \rho_\omega)\big] = -\mathbb{E}\big[\omega^2((Y - \mu)^2 - \sigma^2)(D - e_\omega)(\mu - \rho_\omega)S(O)\big].$$

Collecting all non-zero terms in $\tilde{N}_\omega$, we identify the influence function for the numerator,

$\phi_N(O)$, such that $\tilde{N}_\omega = \mathbb{E}[\phi_N(O)S(O)]$:

$$\begin{aligned}\phi_N(O) &= \big[\omega - \omega^2((Y-\mu)^2 - \sigma^2)\big](D-e_\omega)(\mu-\rho_\omega) + \omega(D-e_\omega)(Y-\mu) - N_\omega \\ &= \big[2\omega - \omega^2(Y-\mu)^2\big](D-e_\omega)(\mu-\rho_\omega) + \omega(D-e_\omega)(Y-\mu) - N_\omega \\ &\equiv A_{aug}(O;\eta_0) - N_\omega,\end{aligned}$$

where we define $\omega_{db}(O) = 2\omega - \omega^2(Y-\mu)^2$.

An identical pathwise differentiation applied to the denominator K_ω yields its influence function. The projection terms identically vanish, leaving:

$$\phi_K(O) = \big[\omega - \omega^2((Y-\mu)^2 - \sigma^2)\big](D-e_\omega)^2 - K_\omega = \omega_{db}(D-e_\omega)^2 - K_\omega \equiv B_{aug}(O;\eta_0) - K_\omega.$$

Finally, combining the components yields the overall unscaled EIF as defined in ψ_{aug}. $\square$

Proof of Theorem 3. Let P denote the true probability measure. To account for cross-fitting, let $\mathcal{I}_k$ denote the observation indices in fold k, let $P_{n,k}$ denote the empirical measure over fold k (i.e., $P_{n,k}[g] = |\mathcal{I}_k|^{-1}\sum_{i\in\mathcal{I}_k} g(O_i)$), and let $P_n = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} P_{n,k}$ denote the full empirical measure. Let $\hat{\eta}_k = (\hat{\omega}_k, \hat{\mu}_k, \hat{e}_k, \hat{\rho}_k)$ denote the nuisance estimators trained on the complement $\mathcal{I}_k^c$. For notational simplicity, we drop the ω subscripts on e and ρ, denote the true parameters as η_0, use $\|\cdot\|_{P,2}$ for the $L_2(P)$ norm, and use $\|\cdot\|_\infty$ for the uniform supremum norm. By Assumption 2(a) and algorithmic trimming, the precision weights ω_0 and $\hat{\omega}_k$ are strictly bounded such that $\|\omega_0\|_\infty \le C$, $\|1/\omega_0\|_\infty \le C$, and $\|\hat{\omega}_k\|_\infty \le C$.

We define the moment functions for the numerator and denominator, corresponding to the components of the augmented score ψ_{aug}, as

$$\begin{aligned}m_A(O;\eta) &= \omega_{db}(O;\eta)(D-e(X))(\mu(D,X)-\rho(X)) + \omega(D,X)(D-e(X))(Y-\mu(D,X)), \\ m_B(O;\eta) &= \omega_{db}(O;\eta)(D-e(X))^2,\end{aligned}$$

where $\omega_{db}(O;\eta) = 2\omega(D,X) - \omega^2(D,X)(Y-\mu(D,X))^2$.

The population parameters are $A_0 = P[m_A(O;\eta_0)]$ and $B_0 = P[m_B(O;\eta_0)]$. Note $B_0 = K_{\omega_0}$ and the optimal estimand is $\tau^*_\omega = A_0/B_0$. The cross-fitted estimators are $\hat{A} = P_n[m_A(O;\hat{\eta}_k)]$ and $\hat{B} = P_n[m_B(O;\hat{\eta}_k)]$.

The true influence functions are $\phi_A(O) = m_A(O; \eta_0) - A_0$ and $\phi_B(O) = m_B(O; \eta_0) - B_0$. We decompose the estimation error for the denominator $\hat{B}$:

$$
\begin{aligned}
\hat{B} - B_0 &= \sum_{k=1}^{K} \frac{|\mathcal{I}_k|}{n}(P_{n,k} - P)[m_B(O; \hat{\eta}_k) - m_B(O; \eta_0)] \\
&\quad + \sum_{k=1}^{K} \frac{|\mathcal{I}_k|}{n} P[m_B(O; \hat{\eta}_k) - m_B(O; \eta_0)] + P_n[\phi_B(O)] \\
&\equiv E_n^B + R_n^B + P_n[\phi_B(O)],
\end{aligned}
$$

where E_n^B is the empirical process term and R_n^B is the remainder bias term. An analogous decomposition $\hat{A} - A_0 = E_n^A + R_n^A + P_n[\phi_A(O)]$ holds for the numerator. We must show that the remainders and empirical process terms are $o_P(n^{-1/2})$ under either Condition A or Condition B.

The Remainder Term R_n^B

We analyze the fold-specific remainder $R_{n,k}^B = P[m_B(O; \hat{\eta}_k) - m_B(O; \eta_0)]$ using the Law of Iterated Expectations conditional on (D, X). Let $\Delta\omega = \hat{\omega}_k - \omega_0$, $\Delta\mu = \hat{\mu}_k - \mu_0$, $\Delta e = \hat{e}_k - e_0$, and $\Delta\rho = \hat{\rho}_k - \rho_0$.

Because $\mathbb{E}[(Y - \hat{\mu}_k)^2 \mid D, X] = \sigma_0^2 + (\mu_0 - \hat{\mu}_k)^2 = 1/\omega_0 + \Delta\mu^2$, the conditional expectation of the debiased weight evaluates to

$$
\begin{aligned}
\mathbb{E}[\omega_{db}(O; \hat{\eta}_k) \mid D, X] &= 2\hat{\omega}_k - \hat{\omega}_k^2 \left(\frac{1}{\omega_0} + \Delta\mu^2 \right) \\
&= 2(\omega_0 + \Delta\omega) - \frac{\omega_0^2 + 2\omega_0\Delta\omega + \Delta\omega^2}{\omega_0} - \hat{\omega}_k^2 \Delta\mu^2 \\
&= \omega_0 - \left(\frac{\Delta\omega^2}{\omega_0} + \hat{\omega}_k^2 \Delta\mu^2 \right) \equiv \omega_0 - \epsilon_{db},
\end{aligned}
$$

where ϵ_{db} captures the second-order estimation errors of the precision weight. Substituting this into the conditional expectation of $m_B(O; \hat{\eta}_k)$ yields

$$
\mathbb{E}[m_B(O; \hat{\eta}_k) \mid D, X] = \omega_0(D - e_0)^2 - 2\omega_0(D - e_0)\Delta e + \omega_0 \Delta e^2 - \epsilon_{db}(D - \hat{e}_k)^2.
$$

The leading term recovers $\mathbb{E}[m_B(O; \eta_0) \mid D, X]$. Taking the unconditional expectation, the first-order error vanishes because $P[-2\omega_0(D - e_0)\Delta e] = -2P[\Delta e \mathbb{E}[\omega_0(D - e_0) \mid X]] = 0$, which confirms Neyman orthogonality. The remainder simplifies to

$$R_{n,k}^B = P[\omega_0 \Delta e^2 - \epsilon_{db}(D - \hat{e}_k)^2].$$

Under Condition A, because all variables and estimators are bounded ($\|\cdot\|_\infty \leq C$), applying the Cauchy-Schwarz inequality yields $|R_{n,k}^B| \leq C\|\Delta e\|_{P,2}^2 + C\|\Delta\omega\|_{P,2}^2 + C\|\Delta\mu\|_{P,2}^2 = o_P(n^{-1/2})$.

Under Condition B, the projection error satisfies $|P[\omega_0 \Delta e^2]| \leq \|\omega_0\|_\infty \|\Delta e\|_{P,2}^2 = o_P(n^{-1/2})$. Notice that $(D - \hat{e}_k) = (D - e_0) - \Delta e$. Thus,

$$(D - \hat{e}_k)^2 \leq 2(D - e_0)^2 + 2\Delta e^2 \leq 2v(O)^2 + 2\Delta e^2.$$

Evaluating the $\Delta\mu^2$ component of ϵ_{db}:

$$\begin{aligned}|P[\hat{\omega}_k^2 \Delta\mu^2 (D - \hat{e}_k)^2]| &\leq 2\|\hat{\omega}_k\|_\infty^2 \left(P[\Delta\mu^2 v(O)^2] + P[\Delta\mu^2 \Delta e^2]\right) \\ &\leq 2\|\hat{\omega}_k\|_\infty^2 \left(\|\Delta\mu \cdot v(O)\|_{P,2}^2 + \|\Delta\mu\|_\infty^2 \|\Delta e\|_{P,2}^2\right).\end{aligned}$$

Condition B(iii) guarantees $\|\Delta\mu \cdot v(O)\|_{P,2}^2 = o_P(n^{-1/2})$. Because Condition B(ii) guarantees $\|\Delta\mu\|_\infty = o_P(1)$, the second term is $o_P(1) \cdot o_P(n^{-1/2}) = o_P(n^{-1/2})$. The $\Delta\omega^2/\omega_0$ component follows identically. Thus, $R_n^B = o_P(n^{-1/2})$.

The Remainder Term R_n^A

Conditioning on (D, X) for the numerator $m_A(O; \hat{\eta}_k)$ and applying the internal cancellation $\hat{\mu}_k - \hat{\rho}_k - \Delta\mu = \mu_0 - \rho_0 - \Delta\rho$, the first-order errors again vanish by orthogonality: $P[-\omega_0(D - e_0)\Delta\rho] = 0$ and $P[-\omega_0 \Delta e(\mu_0 - \rho_0)] = 0$. The remainder evaluates to

$$R_{n,k}^A = P[\omega_0 \Delta e \Delta\rho - \Delta\omega(D - \hat{e}_k)\Delta\mu - \epsilon_{db}(D - \hat{e}_k)(\hat{\mu}_k - \hat{\rho}_k)].$$

Under Condition A, utilizing uniform boundedness, the Cauchy-Schwarz inequality yields bounds of the form $C\|\Delta e\|_{P,2}\|\Delta\rho\|_{P,2}$, which scale at $o_P(n^{-1/2})$.

Under Condition B, for the first term: $|P[\omega_0 \Delta e \Delta\rho]| \leq \|\omega_0\|_\infty \|\Delta e\|_{P,2} \|\Delta\rho\|_{P,2} = o_P(n^{-1/2})$.

For the cross-term, we decompose $(D - \hat{e}_k) = (D - e_0) - \Delta e$. By the triangle inequality,

$|D - \hat{e}_k| \leq |D - e_0| + |\Delta e| \leq v(O) + |\Delta e|$. Applying the triangle and Cauchy-Schwarz inequalities,

$$\begin{aligned} |P[\Delta\omega(D - \hat{e}_k)\Delta\mu]| &\leq P[|\Delta\omega| \cdot v(O) \cdot |\Delta\mu|] + P[|\Delta\omega| \cdot |\Delta e| \cdot |\Delta\mu|] \\ &\leq \|\Delta\omega \cdot v(O)\|_{P,2}\|\Delta\mu\|_{P,2} + \|\Delta\omega\|_\infty\|\Delta e\|_{P,2}\|\Delta\mu\|_{P,2}. \end{aligned}$$

Because $v(O) \geq 1$, the unweighted rate is bounded by $\|\Delta\mu\|_{P,2} \leq \|\Delta\mu \cdot v(O)\|_{P,2}$. Thus, the first term is $o_P(n^{-1/4})o_P(n^{-1/4}) = o_P(n^{-1/2})$. The second term is $o_P(1) \cdot o_P(n^{-1/4}) \cdot o_P(n^{-1/4}) = o_P(n^{-1/2})$.

For the ϵ_{db} components, we apply the triangle inequality to the product

$$|(D - \hat{e}_k)(\hat{\mu}_k - \hat{\rho}_k)| \leq (v(O) + |\Delta e|)(v(O) + |\Delta\mu| + |\Delta\rho|).$$

We can extract the uniform estimation errors outside the expectation to rigorously bound the $\Delta\mu^2$ component unconditionally. Define the uniform bound $C_n = (1 + \|\Delta e\|_\infty)(1 + \|\Delta\mu\|_\infty + \|\Delta\rho\|_\infty) = \mathcal{O}_P(1)$. Because $v(O) \geq 1$, the product is strictly bounded by $C_n v(O)^2$. Thus,

$$|P[\hat{\omega}_k^2\Delta\mu^2(D - \hat{e}_k)(\hat{\mu}_k - \hat{\rho}_k)]| \leq \|\hat{\omega}_k\|_\infty^2 C_n P[\Delta\mu^2 v(O)^2] = \mathcal{O}_P(1)\|\Delta\mu \cdot v(O)\|_{P,2}^2 = o_P(n^{-1/2}).$$

The $\Delta\omega^2/\omega_0$ component follows identically. Thus, $R_n^A = o_P(n^{-1/2})$.

The Empirical Process Terms E_n^A and E_n^B

Because the nuisance estimators $\hat{\eta}_k$ are computed on the independent sample $\mathcal{I}_k^c$, they act as conditionally centered empirical processes. By Lemma 2 of Kennedy et al. (2020), these processes scale as $\mathcal{O}_P(n^{-1/2}\|m_J(O;\hat{\eta}_k) - m_J(O;\eta_0)\|_{P,2})$ for $J \in \{A, B\}$. We must verify that the $L_2(P)$ norm of the estimated score difference is $o_P(1)$.

Under Condition A, because all variables and estimators are uniformly bounded, the mapping $\eta \mapsto m_J(O;\eta)$ has bounded partial derivatives and is therefore globally Lipschitz continuous. Minkowski's inequality ensures $\|m_J(O;\hat{\eta}_k) - m_J(O;\eta_0)\|_{P,2} \leq L\big(\|\Delta\omega\|_{P,2} + \|\Delta\mu\|_{P,2} + \|\Delta e\|_{P,2} + \|\Delta\rho\|_{P,2}\big) = o_P(1)$, where $L > 0$ is the Lipschitz constant.

Under Condition B, the leading first-order discrepancy term in $m_B(O;\eta)$ is proportional to

$\Delta\omega(Y-\mu_0)^2(D-e_0)^2$. We bound its $L_2(P)$ norm as

$$\begin{aligned}\|\Delta\omega(Y-\mu_0)^2(D-e_0)^2\|_{P,2} &\leq \|\Delta\omega\|_\infty \left(\mathbb{E}[(Y-\mu_0)^4(D-e_0)^4]\right)^{1/2}\\ &\leq \|\Delta\omega\|_\infty \left(\mathbb{E}[(Y-\mu_0)^8]\mathbb{E}[(D-e_0)^8]\right)^{1/4}.\end{aligned}$$

Because Condition B(i) guarantees finite moments up to $q \geq 8$, and true conditional expectations inherit bounds from the data via Jensen's inequality (e.g., $\mathbb{E}[\mu_0(X)^8] \leq \mathbb{E}[Y^8]$), these structural expectations are finite constants. Because the uniform estimation error converges as $\|\Delta\omega\|_\infty = o_P(1)$, the entire product vanishes to $o_P(1)$.

Quadratic estimation error terms are handled identically:

$$\|\hat{\omega}_k^2\Delta\mu^2(D-e_0)^2\|_{P,2} \leq \|\hat{\omega}_k\|_\infty^2\|\Delta\mu\|_\infty^2 \left(\mathbb{E}[(D-e_0)^4]\right)^{1/2} = \mathcal{O}_P(1)\cdot o_P(1)\cdot\mathcal{O}(1) = o_P(1).$$

Identical logic applies to all remaining lower-order terms in m_A and m_B. Consequently, under either condition, $E_n^A = o_P(n^{-1/2})$ and $E_n^B = o_P(n^{-1/2})$.

Asymptotic Normality via the Delta Method

We have established that both $\hat{A}$ and $\hat{B}$ are RAL estimators for A_0 and B_0:

$$\begin{aligned}\hat{A}-A_0 &= P_n[\phi_A(O)] + o_P(n^{-1/2}),\\ \hat{B}-B_0 &= P_n[\phi_B(O)] + o_P(n^{-1/2}).\end{aligned}$$

By Assumption 2(b), the global variance is non-degenerate, $B_0 > 0$. Applying the functional Delta method to the ratio $\hat{\tau}_\omega^* = \hat{A}/\hat{B}$ yields:

$$\begin{aligned}\hat{\tau}_\omega^* - \tau_\omega^* &= \frac{1}{B_0}(\hat{A}-A_0) - \frac{A_0}{B_0^2}(\hat{B}-B_0) + o_P(n^{-1/2})\\ &= P_n\left[\frac{\phi_A(O)-\tau_\omega^*\phi_B(O)}{B_0}\right] + o_P(n^{-1/2}).\end{aligned}$$

Substituting the exact expressions for the parameter influence functions:

$$\begin{aligned}\phi_A(O)-\tau_\omega^*\phi_B(O) &= (m_A(O;\eta_0)-A_0) - \tau_\omega^*(m_B(O;\eta_0)-B_0)\\ &= m_A(O;\eta_0) - \tau_\omega^* m_B(O;\eta_0) - (A_0 - \tau_\omega^* B_0).\end{aligned}$$

Since $\tau^*_\omega = A_0/B_0$, the constant term exactly cancels. Recalling the unscaled augmented score ψ_{aug} from Theorem 2, the scaled influence function identically simplifies to $B_0^{-1}\psi_{aug}(O; \tau^*_\omega, \eta_0)$. By the Central Limit Theorem, $\sqrt{n}(\hat{\tau}^*_\omega - \tau^*_\omega) \xrightarrow{d} \mathcal{N}(0, V^*_\omega)$. This completes the proof. $\square$

B Appendix: Extra Results

B.1 Generalization to Categorical Treatments

We now demonstrate how our continuous balancing weights framework incorporates existing results for categorical treatments $D_i \in \{1, \ldots, J\}$ as special cases. In this discrete setting, the analytical focus shifts from a continuous derivative to pairwise contrasts defined relative to a target population specified by a marginal tilting function $w(X_i)$, normalized such that $\mathbb{E}[w(X_i)] = 1$.

Let $e_j(x) = \mathbb{P}(D_i = j|X_i = x)$ be the generalized propensity score and let $\sigma^2(j, x) = Var(Y_i|D_i = j, X_i = x)$ denote the conditional outcome variance. The weighted pairwise contrast between treatment j and k is

$$\tau_{jk}(w) = \mathbb{E}[w(X_i)(\mu(j, X_i) - \mu(k, X_i))].$$

The discrete Riesz Representer (balancing weight) for $\tau_{jk}(w)$ that satisfies the covariate balance constraint from Definition 1 is

$$\alpha_{jk}(D_i, X_i; w) = w(X_i)\left(\frac{\mathbb{I}(D_i = j)}{e_j(X_i)} - \frac{\mathbb{I}(D_i = k)}{e_k(X_i)}\right).$$

Following Equation (5), the corresponding conditional efficiency bound $V_{S,jk}(w)$ is

$$\mathbb{E}[\alpha_{jk}^2(D_i, X_i; w)\sigma^2(D_i, X_i)].$$

We evaluate this expectation by taking the iterated expectation over the discrete support of D_i

$$\begin{aligned}
V_{S,jk}(w) &= \mathbb{E}\left[\sum_{l=1}^{J} e_l(X_i)\alpha_{jk}^2(l, X_i; w)\sigma^2(l, X_i)\right] \\
&= \mathbb{E}\left[w^2(X_i)\left(e_j(X_i)\frac{\sigma^2(j, X_i)}{e_j^2(X_i)} + e_k(X_i)\frac{\sigma^2(k, X_i)}{e_k^2(X_i)}\right)\right] \\
&= \mathbb{E}\left[w^2(X_i)\left(\frac{\sigma^2(j, X_i)}{e_j(X_i)} + \frac{\sigma^2(k, X_i)}{e_k(X_i)}\right)\right].
\end{aligned}$$

With Multiple Treatments ($J \geq 3$). When comparing multiple nominal treatments, Li and Li (2019) proposed selecting the target population by minimizing the total asymptotic variance of all possible pairwise contrasts. We seamlessly adopt this criterion Minimize $\sum_{1 \leq k < j \leq J} V_{S,jk}(w)$ subject to $\mathbb{E}[w(X_i)] = 1$.

$$\sum_{1 \leq k < j \leq J} V_{S,jk}(w) = \mathbb{E}\left[w^2(X_i) \sum_{1 \leq k < j \leq J} \left(\frac{\sigma^2(j, X_i)}{e_j(X_i)} + \frac{\sigma^2(k, X_i)}{e_k(X_i)} \right) \right].$$

In the summation over all unique pairs, each specific term $\sigma^2(l, X_i)/e_l(X_i)$ appears exactly $J - 1$ times. Thus, the objective simplifies to

$$\sum_{1 \leq k < j \leq J} V_{S,jk}(w) = (J-1)\mathbb{E}\left[w^2(X_i) \sum_{j=1}^{J} \frac{\sigma^2(j, X_i)}{e_j(X_i)} \right].$$

Let $\kappa(X_i) = \sum_{j=1}^{J} \sigma^2(j, X_i)/e_j(X_i)$. We seek to minimize $\mathbb{E}[w^2(X_i)\kappa(X_i)]$ subject to $\mathbb{E}[w(X_i)] = 1$. By the Cauchy-Schwarz inequality or a standard Lagrangian approach, the optimal variance-minimizing weight is inversely proportional to $\kappa(x)$

$$w^*(x) \propto \left(\sum_{j=1}^{J} \frac{\sigma^2(j, x)}{e_j(x)} \right)^{-1}.$$

Under homoskedasticity ($\sigma^2(j, x) = \sigma^2$), the outcome variance terms cancel out, and the optimal weight becomes the harmonic mean of the generalized propensity scores, $w^*(x) \propto \left(\sum_{j=1}^{J} 1/e_j(x) \right)^{-1}$, which perfectly recovers the Generalized Overlap Weights of Li and Li (2019).

With Binary Treatment ($J = 2$). If $D_i \in \{0, 1\}$, there is only one relevant contrast, τ_{10}. The optimization criterion naturally reduces to minimizing $V_{S,10}(w)$. Following the derivation above, the optimal target weight is

$$w^*(x) \propto \left(\frac{\sigma^2(1, x)}{e_1(x)} + \frac{\sigma^2(0, x)}{e_0(x)} \right)^{-1}.$$

Under homoskedasticity, this simplifies to $w^*(x) \propto (1/e_1(x) + 1/e_0(x))^{-1} = e_1(x)e_0(x)$. Since $e_1(x) + e_0(x) = 1$, this algebraic equivalence exactly recovers the optimal overlap weights formally established by Crump et al. (2006) and Li et al. (2018).

B.2 Generalization of Theorem 5.1 of Crump et al. (2006)

When estimating an average derivative effect τ_w based on a pre-specified derivative weight $w(d, x)$, the asymptotic variance of an efficient estimator V_ϕ generally decomposes as $V_\phi = V_\psi + V_{adj} + 2C_{\psi,adj}$. Here, V_ψ is the variance assuming the true weights are known, V_{adj} is the variance inflation from estimating the unknown nuisance parameters within the weights, and $C_{\psi,adj}$ is their covariance.

In certain specific settings, structural orthogonality conditions guarantee that the covariance term $C_{\psi,adj}$ is exactly zero. This leads to a simpler, additive decomposition ($V_\phi = V_\psi + V_{adj}$), implying that evaluating the estimand using estimated weights strictly *increases* the asymptotic variance ($V_\phi \geq V_\psi$).

We present a generalized corollary that isolates the exact mathematical conditions required for this simplification to hold, formalizing the findings of Theorem 5.1 in Crump et al. (2006). Let $Z_i = (D_i, X_i)$.

Corollary 1. *Assume that the weight function is defined as $w(Z_i) = \lambda(\eta(W_i))$, where $\lambda(\cdot)$ is a known continuously differentiable mapping, $W_i \subset Z_i$ is a subvector of covariates, and $\eta(w) = \mathbb{E}[H(O_i)|W_i = w]$ for some measurable function $H(O_i) \in L_2(P)$. Furthermore, assume the following two orthogonality conditions hold*

1. $\mathbb{E}[(Y_i - \mu(Z_i))(H(O_i) - \eta(W_i))|Z_i] = 0$.
2. $\mathbb{E}[w(Z_i)(\mu'(Z_i) - \tau_w)\mathbb{E}[\lambda'(\eta(W_i))(\mu'(Z_i) - \tau_w)|W_i](H(O_i) - \eta(W_i))] = 0$.

Under these conditions, the covariance between the fixed-weight influence function $\psi(O_i)$ and the pathwise adjustment term $\phi_{adj}(O_i)$ is identically zero ($C_{\psi,adj} = 0$). Assuming the unnormalized weights are scaled by $K = \mathbb{E}[w(Z_i)]$, the efficiency bound decomposes as $V_\phi = V_\psi + V_{adj}$, where

$$V_{adj} = \frac{1}{K^2}\mathbb{E}\left[\left(\mathbb{E}[\lambda'(\eta(W_i))(\mu'(Z_i) - \tau_w)|W_i]\right)^2 Var(H(O_i)|W_i)\right].$$

Corollary 1 explains why the simple additive variance penalty conventionally holds in the binary treatment setting. In the binary case, target populations depend purely on the propensity score $\eta(X_i) = \mathbb{P}(D_i = 1|X_i)$. Therefore, $W_i = X_i$ and $H(O_i) = D_i$. Condition 1 is trivially satisfied because $(D_i - e(X_i))$ is deterministic given Z_i, yielding $\mathbb{E}[Y_i - \mu(Z_i)|Z_i] = 0$. Condition 2 is satisfied because the binary estimand evaluates the contrast $\Delta\mu(X_i)$, which is independent of D_i. This ensures $\mathbb{E}[D_i - e(X_i)|X_i] = 0$.

This corollary also explains why optimal continuous treatments under general heteroskedasticity preclude this simple additive decomposition. The continuous optimal balancing weights rely on estimating the precision weight $\omega(Z_i)$, which sets $W_i = Z_i$ and $H(O_i) = (Y_i - \mu(Z_i))^2$. In this scenario, Condition 1 evaluates the third conditional moment of the outcome residual

$$\mathbb{E}\left[(Y_i - \mu(Z_i))\left((Y_i - \mu(Z_i))^2 - \sigma^2(Z_i)\right) \mid Z_i\right] = \mathbb{E}[(Y_i - \mu(Z_i))^3|Z_i].$$

Unless the conditional error distribution is symmetric, this conditional covariance is non-zero, violating Condition 1. Furthermore, because the continuous estimand $\mu'(Z_i)$ depends on D_i, Condition 2 is violated when estimating the continuous propensity score $e_\omega(X_i)$.

Proof of Corollary 1. Accounting for the normalization constant $K = \mathbb{E}[w(Z_i)]$, the true influence function assuming fixed weights is

$$\psi(O_i) = \frac{1}{K}\alpha_w(Z_i)(Y_i - \mu(Z_i)) + \frac{w(Z_i)}{K}(\mu'(Z_i) - \tau_w).$$

Under the assumption that the weights depend smoothly on $\eta(W_i) = \mathbb{E}[H(O_i)|W_i]$, standard pathwise differentiation yields the adjustment term

$$\phi_{adj}(O_i) = \frac{1}{K}\mathbb{E}[\lambda'(\eta(W_i))(\mu'(Z_i) - \tau_w)|W_i](H(O_i) - \eta(W_i)).$$

Let $C(W_i) = \frac{1}{K}\mathbb{E}[\lambda'(\eta(W_i))(\mu'(Z_i) - \tau_w)|W_i]$. The covariance $C_{\psi,adj} = \mathbb{E}[\psi(O_i)\phi_{adj}(O_i)]$ expands to

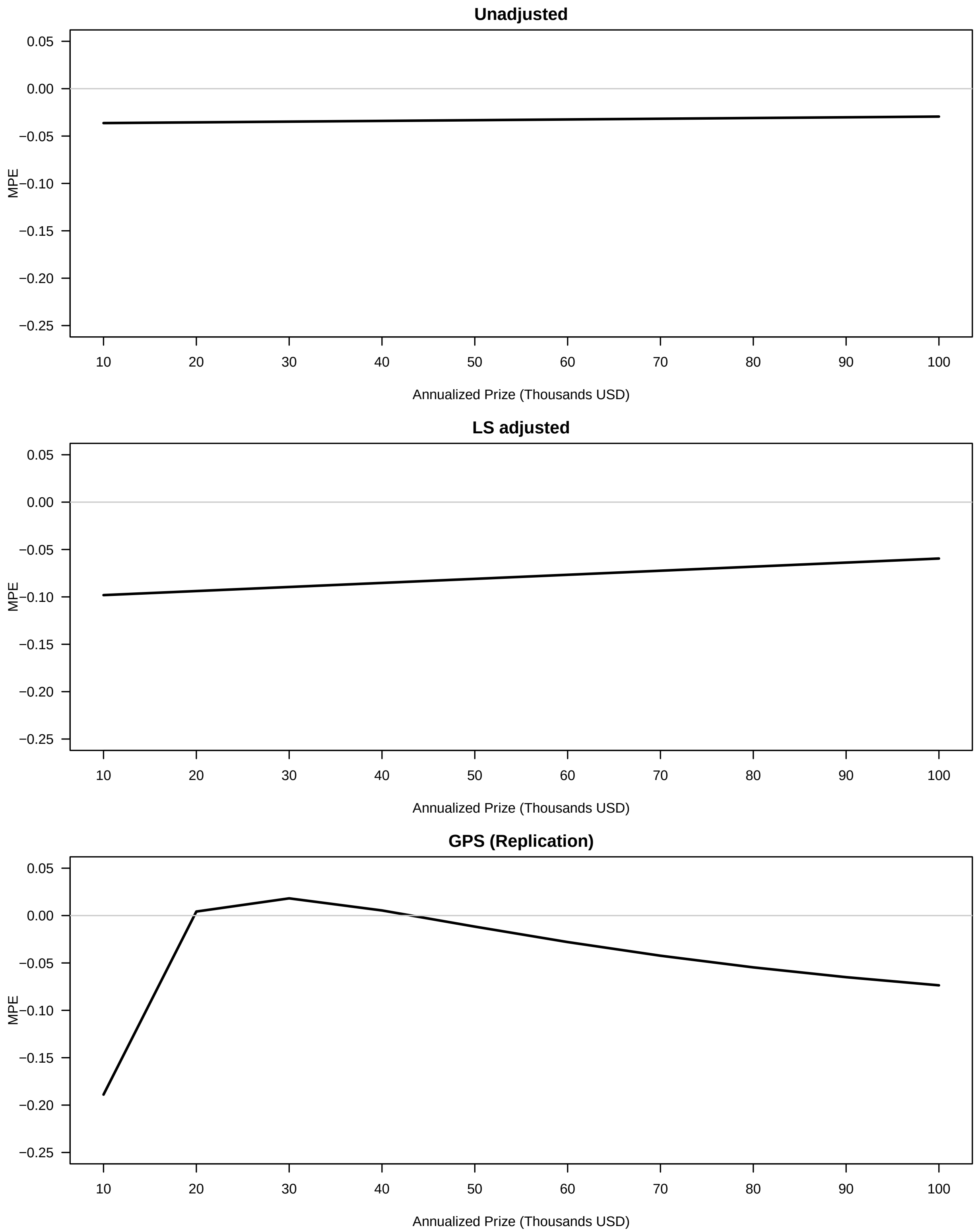
$$\begin{aligned}\mathbb{E}[\psi(O_i)\phi_{adj}(O_i)] = \mathbb{E}&\left[\frac{1}{K}\alpha_w(Z_i)(Y_i - \mu(Z_i))C(W_i)(H(O_i) - \eta(W_i))\right] \\ &+ \mathbb{E}\left[\frac{w(Z_i)}{K}(\mu'(Z_i) - \tau_w)C(W_i)(H(O_i) - \eta(W_i))\right].\end{aligned}$$

By the Law of Iterated Expectations conditioning on Z_i, the first term contains $\mathbb{E}[(Y_i - \mu(Z_i))(H(O_i) - \eta(W_i))|Z_i]$, which vanishes under Condition 1. And the second term vanishes under Condition 2. Consequently, $C_{\psi,adj} = 0$. The variance $V_{adj} = \mathbb{E}[\phi^2_{adj}(O_i)]$ follows directly from $\mathbb{E}[(H(O_i) - \eta(W_i))^2 | W_i] = Var(H(O_i)|W_i)$. $\square$

C Appendix: Replication of HI04

Appendix Table C1: Summary Statistics for Lottery Winners Sample

Variable	Mean	SD
Outcome and Treatment		
Earnings (Y)	10.318	13.163
Annualized Prize (D)	55.196	61.803
Demographics		
Age at Winning	46.945	13.797
Years High School	3.603	1.071
Years College	1.367	1.601
Male	0.578	0.495
Tickets Bought	4.570	3.282
Working at Winning	0.802	0.400
Year Won	1986.059	1.294
Pre-Treatment Earnings		
Earnings year (-6)	11.965	11.790
Earnings year (-5)	12.115	11.992
Earnings year (-4)	12.037	12.081
Earnings year (-3)	12.820	12.654
Earnings year (-2)	13.479	12.965
Earnings year (-1)	14.468	13.624
Observations	237	



Appendix Figure C1: Replication of Figure 7.1 in HI04